MILITARY ENGINEERS SERVICES

NAME OF WORK :CA No: 969/WS - /2024-25 : PROVN OF BOREWELL FOR AUGMENTATION OF WATER SUPPLY AT BANDI AND MOHURA UNDER GE 969 EWS.

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(Signature of Contractor)

AGE (Contracts)
For Accepting Officer

Dated : _____

REGISTERED
Garrison Engineer
969 Engr Works Sec
C/O 56 APO

8103/969/WS- /2024-25/ 40 /E8 20 Aug 2025

M/S _____

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NAME OF WORK :CA No: 969/WS - /2024-25 : PROVN OF BOREWELL FOR AUGMENTATION OF WATER SUPPLY AT BANDI AND MOHURA UNDER GE 969 EWS.

Dear Sir(s),

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in. The tender is on single stage two cover /two state two cover/three cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the **NOTICE INVITING TENDR (NIT).** No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as nonbonafide tender/bid.

3. Bid will be opened on due date and time fixed for opening in the presence of tenders/bidders or their authorized representatives, who have uploaded their quotation bid and who wish to be present at the time of opening the bids.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and or depute your technical representative for discussion on tender/ drawings and to clarify doubts, if any, on or before You are requested not to write piece meal points and forward your points duly consolidated before due date viz

5. Un enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix 'A' to NIT alongwith EARNEST MONEY DEPOSIT (EMD) and tender fee on e-procurement portal and submit the physical documents in the office of GE 969 EWS within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. (a) Contractor having not executed standing security bond and standing security deposit in any MES formation shall upload scanned copy of EARNEST MONEY DEPOSIT(EMD) mentioned in Notice of Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before date & time fixed for this purpose. In case of failure to abide by any of these two requirements, the finance bid will not be opened.

(b) Contractor having not executed standing security bond and standing security deposit in any MES formation would be required to deposit individual security deposit on acceptance of tender which will be calculated with reference to the tendered cost as per scaled laid down by MES for calculation of "EARNEST MONEY" enhanced by 25% subject to maximum of Rs. 1875000/- (Rupees Eighteen Lakhs seventy five thousand Only)

Contd…..

7. Enlisted contractions of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx 'A' to NIT on e-procurement portal and submit physical documents in the office of GE 969 EWS before date & time fixed for this purpose.

8. The contractor must ensure that the tender/bid on the proper from is uploaded in time as the Accepting Office will take no cognizance of any quotations/offer received to any other electronic or physical form like email/ fax/by hand/through post from tenderer/bidder even if they are received in time.

9. In view of delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary, sufficiently in advance of the last due date and time fixed.

10. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of minimum fair wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE concerned and this office.

11. ANY TENDERER, WHICH PROPOSES ALTERNATIONS TO ANY OF THE CONDITION, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

12. Performance Security.

(i) Within 28 days of receipt of the Letter of Acceptance, the successful contractor in shall deliver to the Accepting officer a performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum.

(aa) A Bank Guarantee in the prescribed form.

(bb) Government Securities, FDR or any other Government instruments stipulated by Accepting officer.

(ii) If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

(iii) The period of validity of bank guarantee bond against performance security shall be upto and including Sch of expiry of defect liability period.

(iv) Failure of the successful contractor to comply with the requirements of sub clause12 (i) above shall constitute grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor. Amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

13. REFUND OF PERFORMANCE SECURITY:-The performance security deposit as mentioned as above (as amended vide amendment No 45 to condition 68 of IAFW-2249) shall be refunded to the contractor after the expiration of the defect liability period provide always that the contractor shall first have been paid final bill and have rendered a No-Demand Certificate (I.A.F.W-451). Recovery/withholding due amount if outstanding against the contractor shall be effected from the Performance Security as stipulated under condition 67 of IAFW 2249 (as amended vide amendment No 44)

14. DEFECT LIABILITY PERIOD:-The defect liability period shall be TWENTY FOUR CALENDAR MONTHS after the work has been handed over to Government i.e. after actual date of completion (as per condition 46 of IAFW 2249 amended vide amendment No 48)

Yours faithfully,

_____

Encls:- (As above)

AGE (Contracts)

(Signature of Contractor) For Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. EARNEST MONEY DEPOSIT (EMD)

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed standing security bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, alongwith their tender/bid.

(a) Deposit at Call Receipt from a Scheduled Bank in favour of Garrison Engineer/ CCE concerned.
(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of Garrison Engineer/CCE.

It is advisable that Earnest Money is deposited in the form of deposit call receipt from and approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'EARNEST MONEY DEPOSIT' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE.

NOTES: Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. NON-SUBMISSION OF EARNEST MONEY DEPOSIT (EMD) (Scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER II (FINANCE BID).

2. PERFORMANCE SECURITY

The tenderers shall note that the successful tenderer is required to lodge Performance Security for an amount equal to 3% of the Contract Sum within 28 Days of receipt of Letter of Acceptance in the form of Bank Guarantee in the prescribed format or Government Securities or FDR or any other Government Instruments stipulated by the Accepting Officer in terms of Condition – 19 of IAFW – 2249, General Conditions of Contracts. The Performance Security shall be valid till expiry of Defects Liability Period as stipulated hereinafter. In case the tenderer fails to deposit the Performance Security within the stipulated period, the award of work shall be cancelled and Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of Tender, shall be notified to the tenderer / bidder for depositing the amount through MRO. Bids of such Contractors / bidders shall not be opened till the aforesaid amount equal to the Earnest Money is deposited by him in Government Treasury. The contractor / bidder shall not be entitled for any compensation whatsoever on this account.

3. CONTRACTORS ENLISTED IN MES FORMATION

Contractors whose names are on the approved list of any MES formation and who have deposited Standing Security and have executed Standing Security Bond may tender / bid without depositing Earnest Money with the bid and if the Accepting Officer decides to accept the tender / bid, such tenderers will be required to lodge Performance Security as notified by the Accepting Officer as specified here-in-before.

4. GENERAL INSTRUCTIONS FOR COMPLIANCE

4.1 The bids received only in the electronic form will be considered. All bids shall be submitted on eprocuremes.gov.in portal. Documents should be scanned and forwarded in 'pdf' form and 'xls' form as indicated.
4.2 Bids shall be uploaded on eprocuremes.gov.in portal on or before the bid closing date mentioned in the tender. NO tender/bid in any other electronic or physical form like email/fax/by hand/through post will be considered.
4.3 Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections/ alterations shall be signed /initialed by the lowest bidder after acceptance.
4.4 Drawings, if issued in physical form must be returned duly initialed by the tenderer/bidder in separate envelope indicating his name and address.

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4.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialed. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

4.6 In the technical bid, a scanned copy of Power of Attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid /tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the document confirming of such authority shall be attached with the tender/bid in 'pdf' form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of the firm or company using his DSC shall upload with the tender/bid a scanned copy (in pdf form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to contract including the Arbitration Clause.

4.7 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing his to sign tender in pursuance of which bids are being uploaded by such person as a routine , fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Articles of Association.

4.8 Hard copies of all above documents should be sent by the contractor to the Tender issuing Authority well in advance to be received before the date & time fixed for the same.

4.9 Bid (Cover 1& 2) shall be uploaded online well in time.

4.10 The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24 & 25 of IAFW-2249(General Conditions of contract).

4.11 Tenderers/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointment time.

4.12 The tenderer/bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted; else the bid will be disqualified and summarily rejected.

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4.13 In case the tenderer/bidder has to revise/modify the rates quoted in the BOQ (excel sheet) he can do so only in the BOQ, through eprocuremes.gov.in site only before the bid closing time and date.

4.14 **In case the BOQ is revised by the Department and the bidders has failed to quote in revised BOQ (i.e bidder quoted in previous in BOQ), such bid shall be treated as willful negligence by the bidder and his quotation/bid shall be considered non-bonafied and suitable action deemed fit shall be taken as against the bidder.**

5. **REVOKATION/REVISION OF OFFER UPWARD/ORFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWEST BIDDER**

In the event of lowest tenderer/bidder revoking his offer or revising his rates upward/offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt. treasury. In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

6. **CPM (Critical Path Method)**

6.1 The project planning for work covered in the scope of tender is based on CPM.

6.2 The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/bidder may make use of.

6.3 The tenderer's/bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using eh technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

6.4 Department may issue amendments/errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with all the errata/amendments/corrigendum, if any, issued by the department.

7. These instructions shall form part of the contract documents.

_____
AGE (Contracts)
For Accepting Officer

(Signature of Contractor)

Dated : _____

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work cost more or less, a tenderer/bidder will have no claim on that account. The tender shall be based on as mentioned in aforesaid Appendix 'A'.

3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.

4. Normally contractors whose names are on the MES approved list for the area in which the work lies, and within whose financial category the estimated amount would fall, may tender/bid but in case of term contracts, contractors of categories SS to E may tender/bid. In case, where the tender amount is in excess of the financial limit of the contractor and accepting Officer decides to accept the tender/bid, in which event the tenderer/bidder would be required to lodge additional security deposit as notified by the Accepting Officer in term of conditions of contract. Contractors whose names are on the MES approved list of any MES formation and who have deposited standing security and have executed standing security bond may also tender/bid without depositing Earnest money along with the tender/bid and if the tender/bid submitted by such a tenderer/bidder is accepted, the contractor will be required to lodge with the Controller of Defence Accounts concerned the amount of 'Individual Security deposit' within thirty days of the receipt by him of notification of acceptance of his tender/bid, failing which this sum will be recovered from 1st RAR payment or from the first final bill. In the case of term/running contracts, remaining sum shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.

5. The Office of GE 969 EWS will be the Accepting Officer here in after referred to as such for purpose of the contract.

6. The Technical bid and Financial bid(Cover-1 and Cover-2) shall be uploaded by the tenderer/ bidder on or before the date & time mentioned in NIT. A scanned copy of DD with enlistment details/documents shall be uploaded as packet 1/cover-1 ('T' bid) of the tender/bid on e-tendering portal. DD is refundable in case T bid is not accepted resulting in non-opening of 'Q' Bid. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD and shall not have any claim from Government whatsoever on this account.

6.1. Tender form and conditions of contract and other necessary documents shall be available on eprocuremes.gov.in/ eprocure.gov.in site for download and shall form part of contract agreement in case the tender/bid is accepted.

6.2 In case of contractor who has not executed the Standing Security Bond, the Cover-I shall be accompanied with by Earnest Money of amount as mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CCE/GE/GE(I)/ AGE(I) (see Appendix 'A') by a scheduled Bank or in receipted treasury Challan the amount being credited to the revenue deposit of the concerned CCE/GE/GE(I)/ AGE(I)(see Appendix 'A').

Contd…..

NOTICE INVITING TENDER (NIT)Contd..

6.3 A contractor who is not enlisted for the area in which the work lies but whose name is the MES approved list of any MES formation and who has deposited standing security and executed standing security Bond may bid without depositing earnest money deposit alongwith the tender; but if the Accepting officer accepts the tender/bid, the contractor will be required to lodge with the Controller of Defence Accounts concerted the amount of 'Individual security deposit' within thirty days of the receipt by him of notification of acceptance of his tender/bid, failing which this sum will be recovered from 1st RAR payment or from the first final bill. In the case of term/running contracts, remaining sum shall be recovered from subsequent bill(s) of the contractor.

6.4. A contractor who has executed standing security Bond but not corresponding to the appropriate class as mentioned above, shall lodge with the Accepting Officer, Additional Security Deposit as notified by the Accepting Officer within thirty days of the receipt of his notification of acceptance of his tender/bid, failing which this sum will be recovered from 1st RAR payment or from the first final bill. In the case of term/running contracts, remaining sum shall be recovered from subsequent bill(s) of the contractor. However, in case where any payment is made to the contractor within thirty days of the receipt by him of notification of acceptance of tender/bid, the amount of additional security deposit shall be recovered from such payment.

6.5 The CCE/GE/GE(I)/ AGE(I) will return the Earnest Money wherever applicable to all unsuccessful tenderers/ bidders by endorsing an authority on the deposit-at-call receipt for its refund, on production by the tenderer/bidder a certificate of the Accepting Officer that a bonafide tender/bid was recd and all documents were returned.

6.6 The CCE/GE/GE(I)/ AGE(I) will either return the Earnest Money to the successful tenderer/bidder by endorsing an authority on the deposit-at-call Receipt for its refund on receipt of an appropriate amount of Security Deposit or will retain the same in part or full on account of security deposit if such a transaction is feasible.

6.7 Copies of the drawings and other document pertaining to the work signed for the purpose of identification by the Accepting Officer of his accredited representative, sample of materials and stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting officer and concerned GE/GE(I)/AGE(I) during working hours.

7. The tenderers/bidders are advised to visit the site of work by making prior appointment with GE/GE(I)/AGE(I)/CCE/Project manager who is also the Executing Agency of the work (see appendix 'A'). The tenderers/bidders are deemed to have full knowledge of all relevant documents, samples, site etc, whether they have inspected them or not.

8. Any tender/bid which propose any alteration to any of the conditions laid down or which proposes any other condition or prescription whatsoever, is liable to be rejected.

9. The uploading of bid implies that bidder has read this notice and the Conditions of Contract and has made himself aware of the scope and specification of work to be done and of the conditions and rates at which stores, tools and plants etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

10. Tenders/bidders must be in possession of a copy of the MES Standard Schedule of Rates (see appendix 'A') including amendments and errata thereto.

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NOTICE INVITING TENDER (NIT) Contd..

11. invitation for e-tender does not constitute any guarantee for validation of 'T' bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserve the right to reject the 'T' bid and not open the finance bid of any applicant/bidder. 'T' bid validation shall be decided by the Accepting Officer based on, inter alia, capability of the firm as per criteria given in Appx 'A' to this NIT. The applicant contractor/bidder will be informed regarding non-validation of this 'T' bid assigning reasons thereof through the eprocuremes website. The applicant contractor/bidder if he so desires may appeal to the next higher Engineer authority viz i.e. HQ 133 Works Engineers on email **snr1333-mes@nic.in** with copy to the Accepting Officer on email before the scheduled date of opening of Finance Bid. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The contractor/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

12. (i) It is mandatory on the part of every bidder to be registered with EPFO under Jammu & Kashmir employees provident fund act / scheme 1961. The contractor shall have a EPF code No and all labourers employed by him or sub contractor shall be enrolled as member of provident fund and shall be given universal account Number (UAN).
(ii) Immediately after acceptance of tender, before placing the work order, the contractor shall submit provident Fund code No to the GE.
(iii) Contractor working outside J&K shall also obtain and submit EPF code No of state of J&K as per para (b) above.
(iv) A certificate to effect that all the working labourers employed for execution of work are registered for EPF and due distribution have been credited into their account shall be famished by the contractor alongwith earth running advances or Final Bill

13. The accepting Officer reserves the right to accept a tender submitted by a Public Undertaking, giving a price preference over other Tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderers/bidder whose tender/bid is rejected.

14. Accepting Officer does not bind to accept the lowest or any tender/bid or to give any reason for not doing so.

15. Performance Security.

(i) Within 28 days of receipt of the Letter of Acceptance, the successful contractor in shall deliver to the Accepting officer a performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum.
(aa) A Bank Guarantee in the prescribed form.
(bb) Government Securities, FDR or any other Government instruments stipulated by Accepting officer.

(ii) If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

(iii) The period of validity of bank guarantee bond against performance security shall be upto and including Sch of expiry of defect liability period.

(iv) Failure of the successful contractor to comply with the requirements of sub clause15 (i)above shall constitute grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor. Amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

16. REFUND OF PERFORMANCE SECURITY:-The performance security deposit as mentioned as above (as amended vide amendment No 45 to condition 68 of IAFW-2249) shall be refunded to the contractor after the expiration of the defect liability period provide always that the contractor shall first have been paid final bill and have rendered a No-Demand Certificate (I.A.F.W-451). Recovery/withholding due amount if outstanding against the contractor shall be effected from the Performance Security as stipulated under condition 67 of IAFW 2249 (as amended vide amendment No 44)

17. DEFECT LIABILITY PERIOD:-The defect liability period shall be TWENTY FOUR CALENDAR MONTHS after the work has been handed over to Government i.e. after actual date of completion (as per condition 46 of IAFW 2249 amended vide amendment No 48)

18. This Notice Inviting Tender (NIT) including Appendix 'A' shall from part of the contract.

_____ _____

(Signature of Contractor) AGE (Contracts)
For Accepting Officer

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

MILITARY ENGINEER SERVICES

1	Name of work	**PROVN OF BOREWELL FOR AUGMENTATION OF WATER SUPPLY AT BANDI AND MOHURA UNDER GE 969 EWS**		
2	Estimated Cost	**Rs 9.70 Lakh (At Par Market)**		
3	Period of completion	**270 Days**		
4	Cost of tender documents	Rs 500.00 in the form of DD/Banker's cheque from any Scheduled/Nationalized Bank in favour of GE 969 EWS and payable at Baramulla.(In case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender).		
5	Website/portal address	www.defproc.gov.in		
6	Type of contract	The tender shall be item rate contract based on (IAFW-1779-A) and GCC (IAFW-2249) with Schedule 'A' (BOQ) list of items to be priced by the tenderers electronically .		
7	Information & Details :			
	Description	Date	Time	Remarks
	(a) Bid submission start date	**31 May 2024**	**1000 Hrs**	
	(b) bid submission end date	**06 Jun 2024**	**1800 Hrs**	
	(c) Date of bid opening	**12 Jun 2024 or on subsequent days**	**1000 Hrs**	
8	Eligibility Criteria	**Class'E' Category a(ii)**		
	(A) For MES enlisted contractor	Contractor shall be enlisted with MES in class **'E'** and above and category **a(ii)** subject to the remarks wrt performance in respect of wks in hand reflected in workload return (WLR) circulated by competent engineer authority. Refer Note _ below.		
	(B) For contractors not enlisted with MES	(a) Contractor not enlisted with MES should meet the enlistment criteria of 'E' class & **a(ii)** category contractor with regard to satisfactorily completion of requisite value works, with Central/ State Government/ Central/ State PSUs/AWHO/ AFNHB/CGEWHO/DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in). (b) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (c) Not suspended /debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department or any Central/ State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date. (d) Un-enlisted contractor who have secured two works in MES should get themselves registered in the appropriate designated class with any registering authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES.		
	(C) Technical PQC criteria	In case of Solar Power Works and Runway/Pavements, technical PQC criteria as stipulated in Manual on Contracts and as per policy letters issued by E-in-C's Branch from time to time shall be included suitably.		
9	Tender issuing and Accepting Officer	**Name : Major Niraj Kumar Singh** **Address : GE 969 EWS C/O 56 APO** **e-mail : srew9694-mes@nic.in Phone No : 01952-234271**		
10	Executing agency	GE 969 EWS C/O 56 APO		
11	Earnest Money	**Rs 19,400/-** in favour of GE 969 EWS C/O 56 APO in the form of Deposit at call receipt, FDR not acceptable.		

NOTES.

1. In case after opening of Cover 1, if the number of MES enlisted contractors of eligible class of as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT is less than 7(Seven), applications in respect of MES contractors of one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command/ ADG below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore MES contractor's one class below (two classes below in case of remote and difficult areas) may also bid for this tender. Such contractors (Contractors of one / two class below the eligible class) shall not be considered in case their present residual work in hand is more than FIVE TIMES their present tendering limit. However in case such contractors fulfill the criteria of up gradation to the stipulated eligible class based on past experience of completed works (individual work experience and /or average annual turnover, as applicable) and financial soundness (solvency/financial soundness and working capital), the ceiling of present residual work will not apply and they will be consider for issue of tender. Such bidders shall upload in their Cover-1 bid details related to residual work in hand like details of works in hand showing names of work, names of Accepting officers, contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfill the criteria of up gradation shall also upload the requisite information/ documents in support of up gradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contracts are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class of as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7(Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (individual work experience and/ or average annual turnover as applicable) and financial soundness (solvency/financial and working capital) as per details given in Manual on Contracts. Therefore, such contractors shall upload the requisite information/ documents in the Cover-1.

3. Unenlisted contractor shall considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national/ Indian nationals staying abroad/ Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover 1 for checking eligibility:-

(a) Application for tender on Firm's letter head.

(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vougue.

(c) Scanned copy of DD/ Bankers Cheque toward cost of the tender and EMD instrument in case SSD bond is not signed at the time of registration

(d) Any other document required as described in this Appendix.

(e) Scanned copy of GSTIN Registration and latest clearance certificate from GST-3B i.e, latest clearance certificate form GST-3B of preceding Quarter/ preceding month to the issue of NIT.
(The GST-3B for Last quarter (Quarterly return) and previous month for (Monthly return) should be uploaded, irrespective of date of Tender i.e, month of May 2024/ last Quarter (Jan-Mar 2024)).

5. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility.

(a) Application for tender on Firm's letter head.

(b) Scanned copy of DD/ Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

(c) Copy of Police Verification Report/ Police Clearance Certificate/ Character Certificate for the Police Authority of the area where the registered office of the firm is located/ notarized copy of valid passport of proprietor/ each

partner/ each Director.

(d) All documents required from enlistment in MES for the class mentioned in Para 8(b) above as per Para 1.5 of section 1 of MES Manual on Contracts 2020.

(e) Details of works being executed In MES , if any.

(f) Any other document required as described in this Appendix.

6. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover 1 shall not be considered for validation of 'T' bid and their Financial bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting officer with in 07 days of bid submission end date failing which following action shall be taken.

(a) In case of tender from an enlisted contractor of MES, Where scanned copies of requisite DD/ Bankers Cheque towards cost of tender have been uploaded in cover 1 but physical copies are not received with in the stipulated period, their financial bids (Cover2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding from a period six months commencing from the date of opening of Financial Bid (Cover 2).

(b) In case of tenders from unenlisted contractor, where scanned copies of requisite DD/ Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).

(c) In case of tenders from enlisted and unenlisted contractors where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/ spouse of Proprietor/ Partner/Director and firm's own employees, director, project manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

9. After opening of Cover 1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/ bidders merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/ bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tender/ bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority i.e. HQ 133 Works Engineers on email **snr1333-mes@nic.in** . On emailed with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2 NHEA shall decide the matter within a

period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/ bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an un-enlisted contractor is already executing woks in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of contractor shall be uploaded in the Cover 1 of the bid and shall be checked/ verified by the Accepting Officer.

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/ bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tender after opening of Cover 2 Shall be considered as a wilful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an unlisted tender, Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tender in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/ administrative action shall be taken against such tenders. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/ one or more Partners/ Directors are common. Decision of Accepting Officer on issue/ deny the tender to any one of the related firms shall be final and binding.

15. EPF ACCOUNT NO

(i) Immediately after acceptance of tender, before placing the work order, the Contractor shall submit Provident Fund Code No to the GE.

(ii) A certificate to effect that all working labourers employed for execution of work are registered for EPF & ESIC and due distribution have been credited into their account, shall be furnished by contractor along with each running advances or final bill.

16. DEFECT LIABILITY PERIOD

The work will have defect liability period of 24 calendar months from the date of handing over of asset to government.

Case No : 8103/969/WS- /2023-24/ 03 /E8

AGE (Contracts)
For Accepting Officer

Garrison Engineer
969 Engineer Works Section
PIN-913969
C/o 56 APO
Dated :- 17 May 2024

Amendment to Appendix 'A' To Notice Inviting Tender (NIT)

Ser No.	Page No.	Location /Reference	Particulars of amendments
1	2	3	4
1	11	(i) Para 7 Information & Detail	
		(a) Bid submission start date	For "31 May 2024" (1000 Hrs) Read " 30 Aug 2025" (1000 Hrs)
		(b) Bid submission end date	For " 06 Jun 2024" (1800 Hrs) Read "04 Sep 2025" (1800 Hrs)
		(c) Date of bid opening	For "12 Jun 2024" or on subsequent days (1200 Hrs) Read "06 Sep 2025" or on subsequent days (1000 Hrs)

Signature of Contractor
Dated:

AGE (Contracts)
for GE 969 EWS

IN LIEU OF IAFW-1779-A(REVISED)
(TO BE USED AND READ IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACT IAFW-2249 (1989 PRINT))
MILITARY ENGINEERS SERVICES

REGISTERED
Garrison Engineer
969 Engr Works Sec
C/O 56 APO

8103/969/WS- /2024-25/ /E8 20 Aug 2025

ITEM RATE TENDER AND CONTRACT FOR: PROVN OF BOREWELL FOR AUGMENTATION OF WATER SUPPLY AT BANDI AND MOHURA UNDER GE 969 EWS..

1. Sarvashri/Shri _____of _____ is/are hereby authorized to tender for the subject

work. The tender is to be uploaded on E- DEFPROC.GOV.IN WEB SITE. Bid shall be submitted on line only by the bidders on or before Bid submission End date & time by **1800** hrs on **04 Sep 2025**. Bid/ E-Tender will be opened by bid opening officers as per E-tendering process on bid opening date/time mentioned in bid.

2. All correspondence concerning this tender shall be addressed as indicated at the top of this sheet, quoting the reference as given. For better communication clarification may be addressed on line.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY OTHER TENDER

(Signature of the Officer)
issuing the tender documents
Appointment – AGE(Contracts)
(For and on behalf of the President of India)

_____
Signature of contractor
Dated …….

SCHEDULE 'A'

NOTES :-

1. **SCOPE OF WORK** : The work under this contract broadly consist of **: "PROVN OF BOREWELL FOR AUGMENTATION OF WATER SUPPLY AT BANDI AND MOHURA UNDER GE 969 EWS."** as detailed in Schedule 'A' (List of items of work) complete all as specified. This work comprising of following jobs:-

(i) Job No Minor Wk/02/Fy/2024-25/Rampur : Provn of Borewell for Augmentation of water sup at Mohura

(ii) **Job No Minor Wk/01/Fy/2024-25/Rampur :** Provn of Borwell for Augmentation of water sup at Bandi for the Fy 2024-25.

2. The entire work covered under this contract shall be completed within 270 days from the date of handing over of site given in work order No 1 to be issued by the GE within 7 days of acceptance of tender or after submission of PS by the Contractor.

3. The quoted sum shall be arrived at by the bidder after deducting amount of Sch of Credit from the total amount of the BOQ.

4. The quantities given in BOQ under column 4 are provisional and are inserted as a guide only. These shall however not be varied beyond the limits laid down in condition 7 of IAFW-2249 (General conditions of contracts).

5. The Rates in figures only are to be inserted by bidders on line only in column 13 of Schedule 'A'. The rates in words/amount in figures under column 54/55 will be automatically appear after enable. In case of discrepancy rate in word shall be prevail.

6. Unit rates of all items mentioned in Schedule 'A' shall be inclusive of supply and fix, material and labour, laying, jointing, commissioning, testing etc, (as applicable) unless mentioned specifically as fixing only or supply only etc.

7. Rates quoted for a particular item and/or lump sum quoted by the tenderer shall be deemed to include for any minor details/items of work or constructions which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of the work. Decision of the Accepting Officer as to whether any minor details of work and/or construction is obviously and fairly intended to be included in the contract or not shall be final, conclusive and binding.

8. The description of items of Sch 'A' are brief. These are deemed to be amplified and read in conjunction with special condition, particular specification for material and workmanship, conditions and pre-ambles given in relevant trade sections of MES standard schedule of rates Part-I (2009) and Part-II (2020) and contract drawings/including notes thereon (if attached). The provisions given in the description of item and/or given in particular specification found at variance if any with the provisions contained conditions/preambles to MES Schedule item, the provisions contained in description of item in Sch 'A' and/or in particular specifications, shall take precedence.

9. In case different rates are quoted for the items of similar nature of works, the lowest rate quoted amongst these items shall be applicable for other items also for the purpose of payment. No claim whatsoever on this account shall be admissible.

10. Old material obtained from dismantling/demolition to the extent mentioned in Schedule of credit shall become the property of the contractor and recovery shall be made at the recovery rates mentioned in Credit Schedule. The serviceable material not covered in Credit Schedule retrieved if any shall be the property of the Govt and shall be deposited by the contractor in the store yard of the GE without any extra cost. All rubbish and un-serviceable material shall be removed from the site by the contractor and filled in low lying area outside cantonment land as directed by Engineer-in-Charge without any extra cost.

Contd……/-

11. Site for execution of work shall be available as soon as the work is awarded, in case it is not possible for the Deptt. To make entire site available on the award of Contract, the contractor will have to arrange his working programme accordingly. No claim whatsoever for not giving him the entire site on award of Contract and for giving the site gradually will be admissible.

12. Preparatory works/cuttings/leaving/forming chases, holes etc, through /in walls, floors and ceiling as required and making good to match with the adjoining surfaces required in connection with the work included in Schedule `A'. No adjustment shall however be made on the quoted price on this account while pricing of any deviation for items in respect of Schedule `A' and also due to any variation in quantities indicated as `PROVISIONAL' in the tender documents.

13. Wherever damages are caused to the existing road for taking sewage line/water supply lines across the road, the same shall be made good to match with the original surface by the contractor under his own arrangement and without extra cost to Govt.

14. The tenderer shall ensure that no damage is caused to the existing structures/buildings not included in repair. Demolition process shall be carried out carefully after taking all safety precautions and measures specified in IAFW-2249 without any extra cost to the Govt.

15. The tenderer shall ensure visit to the site for verifying the actual scope of work i.e. quantity/quality of material involved in demolition/dismantling. Whether or not the tenderer actually visited/inspected the site for verifying the condition of the buildings/structures/electrical items to be dismantled/demolished, tenderer shall be deemed to have visited the site and inspected the buildings and has arrived at the scope of dismantling/demolition. No claim, what-so-ever, on this account from the contractor shall be entertained.

16. Electric service connection are to be disconnected by the contractor after taking proper shut down from MES authority prior to start demolition of building. Accordingly water connections from rising main are to be removed and plugged properly. The rates are deemed to be included in the building work.

17. Tenderers to note that old retrieved materials shall not be allowed to be reused in the work unless otherwise specified.

18. TD drawings have not been made part of the tender. However during execution and also before quoting the tender the tenderers can refer these drawings from CWE/GE office during any working day. Contractor shall deem to have studied the drawings before quoting the tender and he shall have no claim whatsoever on this ground.

19. EPF Code & UAN:- Under the Employees Provident Funds and Miscellaneous Provisions Act, 1952, the contractor shall have a EPF Code No and all labours employed by him or sub-contractor shall be enrolled as member of Provident Fund and should be given Universal Account Number (UAN).

20. As per Government of India, Ministry of Labour and Employment New Delhi circular No 35025/02/2014-SS II dated 15 Jul 2015, it is mandatory to the contractor that all the workers deployed by him are enrolled as members of provident fund and shall have the Universal Account Number (UAN).

21. It is mandatory the GE that while clearing the bills/payments (RAR) of contractors, a certificate be obtained from the contractor that all workers employed by him directly or indirectly are registered for EPF and the due contributions have been credited into their accounts.

22. If ISI marked materials are being manufactured, the same shall be ISI marked. For list of ISI marked manufacturers refer website of BIS i.e www.bis.org.in.

23. Execution of work through Power of Attorney holder shall not be allowed.

24. The contractor's quoted rates shall be deemed to be inclusive of all taxes / cesses viz GST duties, Royalties, Octroi & other levies payable under the respective Statutes as applicable on the date of receipt of tender. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as conditional tender and the same shall be liable to be rejected.

25. Performance Security.

(i) Within 28 days of receipt of the Letter of Acceptance, the successful contractor in shall deliver to the Accepting officer a performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum.

(aa) A Bank Guarantee in the prescribed form.

(bb) Government Securities, FDR or any other Government instruments stipulated by Accepting officer.

(ii) If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

(iii) The period of validity of bank guarantee bond against performance security shall be upto and including Sch of expiry of defect liability period.

(iv) Failure of the successful contractor to comply with the requirements of sub clause12 (i) above shall constitute grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor. Amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

26. REFUND OF PERFORMANCE SECURITY:-The performance security deposit as mentioned as above (as amended vide amendment No 45 to condition 68 of IAFW-2249) shall be refunded to the contractor after the expiration of the defect liability period provide always that the contractor shall first have been paid final bill and have rendered a No-Demand Certificate (I.A.F.W-451). Recovery/withholding due amount if outstanding against the contractor shall be effected from the Performance Security as stipulated under condition 67 of IAFW 2249 (as amended vide amendment No 44)

27. DEFECT LIABILITY PERIOD:-The defect liability period shall be TWENTY FOUR CALENDAR MONTHS after the work has been handed over to Government i.e. after actual date of completion (as per condition 46 of IAFW 2249 amended vide amendment No 48)

28. TUBE WELL BORING:-

The depth of boring of the tube well at the site indicated by the GE is anticipated to be 160 meters below ground level. The depth may, however, be ordered in writing to be varied by the GE upto a maximum of 40 meters (plus or minus). If the GE decides that suitable water bearing strata is not available in the aforesaid bore or the boring cannot be continued further on account of the type of strata met with, he may order the contractor to bore at other site (s) selected by the GE, until suitable water bearing strata is met with, or boring cannot continue further on account of the type of strata met with provided the total number of bores (including the first bore) does not exceed 03. The decision of the GE shall be final and binding as to whether or not the bore can be continued further in the strata met with.

SCHEDULE'A'

PART-II (CREDIT SCHEDULE)

Ser No.	Description of items	Unit	Appx No.of Units required	Unit Rate Rs.	Amount Rs.
1	2	3	4	5	6

Notes

1. All quantities in this schedule are 'PROVISIONAL'
2. The tenderer shall visit the site of work to assess the nature and extent of work involved in demolition /dismantling and the quantity of unserviceable material that may be obtained there from. The tenderer shall be deemed to have visited the site and availed the pre-tender opportunity to assess all the eventualities which may affect his quotation whether he actually visit the site or not.
3. The rates for demolition and dismantling under relevant items of Schedule 'A' Part-I shall interalia include for separation of serviceable materials from the unserviceable, stacking the serviceable material within 100 m and disposal of unserviceable material. The decision of the Engineer-in-Charge as to which materials are serviceable or otherwise shall be final and binding. The Unserviceable material shall be neatly stacked at site and the contractor shall take all precaution to protect the unserviceable material from damage from any cause what so ever until the material is allowed to be removed.
4. The unserviceable material obtained under this schedule shall be handed over to the contractor by the EIC on a proper gate pass and credit for demolished/dismantled material for these items will be deducted form the RAR payable subsequent to demolition/ dismantling of the work catered for is executed. But in no case the material shall be allowed to be removed from the site until the credit has been recovered by the Govt.
5. The contractor shall remove the demolished/dismantled material, (other than re-used in the work) from the site of work without any extra cost to the Govt.
6. Material obtained from demolition/dismantling, other than those listed here-in-before and the serviceable material (other than reused in the work) shall be the property of the Govt and shall be removed and deposited in the MES Store Yard without any extra cost of the Govt.

AGE(Contracts)

Signature of Contractor For Accepting Officer

SCHEDULE 'B'

LIST OF MATERIAL TO BE ISSUED TO THE CONTRACTOR

(See condition 10 of IAFW-2249)

Ser No	Description of item	Rate at which materials etc will be issued to the contractor Unit Rate	Place of issue by Name	Remarks
1	2	3 4	5	6

….NIL….

SCHEDULE 'C'

ISSUE OF TOOLS AND PLANTS OTHER THAN TRANSPORT WHICH WILL BE HIRED : REFER CONDITION 15, 34 AND 35 OF IAFW-2249 AND AMENDMENTS THEREOF

Sr No	Quantity	Particulars	Details of MES Crew	Hire Charges per unit per working day	Stand by Charges per unit per off day	Place of Issue by name	Remarks
1	2	3	4	5	6	7	8

_____NIL_____

SCHEDULE 'D'

TRANSPORT TO BE HIRED TO THE CONTRACTOR /REFER CONDITION

(15 AND 35 OF IAFW-2249 OF 1989 PRINT)

S/No	Quantity	Particular	Rate per unit per working day of 8 hours	Place (s) of issue by name	Remarks
1	2	3	4	5	6

_____NIL_____

_____

(Signature of Contractor) AGE(Contracts)

GENERAL SUMMARY

1	Total amount quoted by the contractor against Sch 'A' Part-I brought fwd from Srl page No.	Rs_____
	DEDUCT	
2	**Schedule of credit**	
	Deduct credit for the material to be retrieved from dismantling/demolition/ taking down as listed in Schedule of Credit (Sch 'A' Part -II) brought forward from Srl page No.	**Rs (-)**
3	Net Total C/O to next page.	= Rs

_____
SIGNATURE OF CONTRACTOR
DATED: _____

AGE(Contracts)
FOR ACCEPTING OFFICER

Contd….

T E N D E R

THE PRESIDENT OF INDIA

Having examined and perused the following documents :-

1. Specifications signed by Assistant Garrison Engineer(Contracts)
2. Drawings detailed in the list of drawings.
3. Schedule 'A' 'B' 'C' and 'D' attached hereto.

4. MES STANDARD SCHEDULE OF RATES 2009 (PART-I) SPECIFICATIONS AND MES STANDARD SCHEDULE OF RATES-2020 (PART-II) RATES TOGETHER WITH ERRATA/AMENDMENTS AS FOLLWS:

Part-I
(a) Errata/Amendments 1 to 3
Part-II
(a) Errata/Amendments 1 to 122

5. General condition of contract IAFW-2249 (1989 print) together with the amendment No 1 to **49** and errata 1 to 20. In case of difference in interpretation due to wording of English and Hindi version, English version will prevail.

6. **WATER:** Condition 31 of IAFW-2249 (General Conditions of Contracts). Water will not be supplied by the MES.

7. **Should this tender be accepted I/*We agree**

**(a) That the sum of Rs _____(Rupees_____ forwarded as earnest money shall be forfeited in full and shall be credited into consolidated fund of India on non-submission of performance security deposit as defined in the tender documents.

(b) To execute all the works referred in the said documents upon the terms and condition contained or referred to therein and detailed in items of Schedule 'A' and to carry out such deviations as may be ordered under condition 7 of IAFW-2249 upto maximum of **20% (Twenty percent**) and further agree to refer to all disputes as required vide condition 70 of IAFW-2249 to the sole Arbitration of a serving officer having degree in Engineering of equivalent or having passed final/direct final Examination of Sub-Division-II from Institution of Surveyors (India) recognized by the Govt of India to be appointed by the Chief Engineer 31 Zone, c/o 56 APO or in his absence, the officer officiating as Chief Engineer 31 Zone, c/o 56 APO, if specifically delegated in writing by Chief Engineer 31 Zone, c/o 56 APO whose decision shall be final, conclusive and binding.

** Delete where not applicable.
* Delete whichever is not applicable.

(Contd….)

TOTAL BROUGHT FORWARD FROM SERIAL PAGE NO_____

Rs_____(Rupees_____

_____)

Signature_____(Name _____) in the Capacity

of _____ duly authorized to sign the Tender for and on behalf of

M/S _____ (in block letters)

Postal Address_____ Witness _____

_____ Address _____

_____ _____

Telephone_____ _____

ACCEPTANCE

_____alterations have been made in these documents and as

evidence that these alterations were made before execution of the contract, the same have been

4. initialed by the contractor and Assistant Garrison Engineer(Contracts).

The said Officer is hereby authorized to sign and initial on my behalf the documents forming part of the contract.

The above Tender is accepted by me on behalf of the President of India at the Item

rates contained in Schedule 'A' for Rs _____ (Rupees

On the_____ day of_____ 2024

Signature dates this _____ day of _____2024

Appointment :Garrison Engineer
969 Engr Wks Sec
C/O 56 APO
ACCEPTING OFFICER
(For and on behalf of the President of India)

SPECIAL CONDITIONS

1. **GENERAL**

1.1 The following special conditions shall be read in conjunction with General Conditions of contracts IAFW-2249 (1989 Print), including errata/amendments thereto. If there are any provisions in these special conditions, which are at variance with the provisions in the above-mentioned documents; the provisions in these special conditions shall be deemed to take precedence.

1.2 General Rules, preambles, special conditions, method of measurements etc. specified in MES Standard Schedule of Rates shall be applicable unless specific provisions to the contrary are made in these tender documents.

2.0 VISIT TO SITE (S) OF WORK

The tenderers are advised to visit the site(s) of work with prior permission of GE and make themselves familiar with the local working conditions, accessibility to sites(s), availability of material and other relevant conditions which effect their pricing, workman like execution and entire completion of work. The tenderer shall be deemed to have visited the site(s) before tendering whether they actually visit or not.

3. **RESTRICTION FOR ENTRY TO WORKS SITE**: All works lies in **RESTRICTED AREA.**The restrictions for entry to work site and conditions of working in restricted area shall be as under.

(a) **ENTRY AND EXIT**: The contractor/his agents/representatives/ workmen etc, and his materials, carts, trucks or other means of transport etc. will be allowed to enter through and leave from only such gate or gates and at such times as the GE or authorities in charge of the restricted area may at their sole discretion permit to be used. Contractor's authorised representative is required to be present at the places of entry and exit for purpose of identifying his carts, trucks, etc. to the personal in charge of the security of the restricted area.

(b) **IDENTITY OF WORKMEN :** Every workman shall be in possession of an identity card. The identity card shall be issued after a thorough investigation of antecedents of the labourers by the contractor and attested by Officer-in-Charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for conduct of his workmen, agents or representatives.

(c) **IDENTITY CARD OR PASSES:** The contractor, his agent and representatives are required individually to be in possession of an identity card or pass which will be examined by the security staff at the time of entry into or exit from the restricted area at any time or number of times inside restricted area.

(d) **SEARCH**: Thorough search of all persons and transport shall be carried out at each gate and for as many times gate is used for entry or exit and may also be carried out any number of times at the site within the restricted area.

(e) **WORKING HOURS**: The units controlling restricted area, usually work during six days in a week and remain closed on the 7th day. The working hours available to the contractor's labour and staff are however appreciately reduced because of the time of entry and exit during working hours. The exact working hours, working days and number of working days observed for these restricted area (s) where works are to be carried out shall be deemed to have been ascertained by contractor before submitting his tender. The tenderers' attention is invited to the fact that number of working hours for a unit are prescribed in regulations and that they cannot be increased by the Garrison Engineer or authorities controlling the restricted area. The definition of "working days" as given under condition 1(t) of IAFW-2249 does not apply in case where the works are carried out in restricted area.

(f) **WORKING ON HOLIDAYS** : The contractor shall not carry out any work on gazetted holidays, weekly holidays and other non working days except when he is specially authorised in writing to do so by the Garrison Engineer. The GE may at his sole discretion declare any day as holidays or non working day without assigning any reasons for such declaration.

(g) **FIRE PRECAUTIONS**:

(i) The contractor, his agents, representatives, workmen etc. shall strictly observe the order pertaining to fire precautions prevailing within the restricted area.

(ii) Motor transport vehicle, if allowed by the authorities to enter the restricted area must be fitted with the serviceable fire extinguisher.

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(h) **FEMALE SEARCHER**: If the contractor desires to employ female labour on works to be carried out inside the area of factory, depot, park etc, and a female searcher is not borne on the authorised strength of the factory, depot, park etc. at the time of submission of tender, he shall be deemed to have allowed in his tender for pay and allowances etc for a female searcher (Class IV servant) calculated for the period female labourers employed by him inside that area. If more than one contractor employs female labourers during any month and female's searcher (s) has/have to be employed in addition to the authorised strength of the factory, depot, park etc. the salary and allowances paid to the additional female searchers shall be distributed on equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE's decision in regard to the amount payable on this account by any contractor shall be final and binding.

4.0 **CONTRACTOR'S REPRESENTATIVES/AGENTS AND WORKERS**

4.1 The contractor shall employ only Indian Nationals and verify their antecedents and loyalty before employing them on the work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the work. If for reasons of technical collaboration or other consideration the employment of any foreign national is unavoidable, the tenderers shall furnish full particulars of such persons to the Accepting Officer at the time of submission of tender. As a proof that he has employed only Indian Nationals, the contractor shall render a certificate to this effect to GE within one month from the date of commencement of work.

4.2 The GE shall have full power, and without giving any reasons to order the contractor immediately to cease to employ in connection with this contract any representatives, agent, servants and workmen or employees whose continued employment in his (GE's) opinion is undesirable. The contractor shall not be allowed any compensation on this account.

4.3 Tenderer's attention is also drawn to condition 25 of IAFW-2249 in this connection.

4.4 **VERIFICATION OF ANTECEDENTS:** Verification of antecedents of the Contractor's representatives/ labour deployed at site in connection with execution of work under the Contract, as per security requirement of user unit/ installation shall be the responsibility of the Contractor and all expenses in connection with verification of antecedents by police authority/ security agency shall be born by the Contractor.

5.0 **HOURS OF WORKING**

The working hours shall be as notified by the authority controlling the area, which will be normally from 0800hours to 1700 hours. If, however, contractor wants to work beyond these normal hours he can do so if permitted by the GE in writing. Similarly no work will be allowed to be carried out on Sunday and gazetted holidays unless specifically permitted by the GE in writing.

6.0 CONTRACTOR'S PLANT/ EQUIPMENT AT SITE

6.1 The contractor shall furnish the Engineer-in-Charge every morning a distribution return of the plant and/or equipment on the site of work, stating the following particulars:-

(i) Particulars of plant/or equipment, its make, year of manufacture, Model No. if any registration No, if any, capacity, year of purchase etc.

(ii) Total Nos. (quantity) on site of works.

(iii) Location indicating Nos. (quantity) at each location on the site of works.

(iv) Purchase value on the date of purchase. For the purpose of this condition plant and/or equipment shall include vehicle i.e, lorries but neither the workmen's tools nor any manually operated tools/equipment.

6.2 The Engineer-in-charge shall record the particulars of plant/equipment supplied by the contractor in the works diary and send the return to the GE for record in his office.

7.0 LOSS OR DAMAGE ON ACCOUNT OF ENEMY ACTION

If as a result of enemy action the contractor suffers any loss or damage, the Govt., shall reimburse to the contractor such loss or damage, to the extent and in the manner here-in-after provided:-

SPECIAL CONDITIONS

(i) The loss suffered by him on account of any damage or destruction of his plant and or equipment (as defined in special condition 7 above) or materials or any part or parts thereof. (The amount of loss assessed by the Accepting Officer of the contract or CWE in the case of contract accepted by GE, shall be final and binding).

(ii) The compensation paid by him under any law for the time being in force to any workman employed by him for any injury caused to him or to the Workmen's legal successors for loss of the workman's life.

(iii) Payment of compensation for loss or damage to any work or part of work carried out. The amount of compensation shall be determined in accordance with condition 48 of General Conditions of Contracts (IAFW-2249).

(iv) No reimbursement shall be made nor any compensation be payable under the above provision unless the contractor has taken air defence/precautions ordered in writing by the GE concerned or in the absence of such reasonable precautions. Neither reimbursement shall be payable nor any compensation shall be payable for any plant and/ or equipment or material not lying at site of work at the time of enemy action.

8.0 USE OF EXISTING BUILDINGS ETC.

The contractor shall not use the existing latrines, baths and other existing buildings if any for use by his work people or for storage of his materials or for any other purpose unless permitted to do so in writing by Garrison Engineer.

9. 0 WATER :

Water shall not be supplied by MES. The tenderers are advised to visit the site of works to ascertain availability of water from civil sources or from nearby natural sources out side MOD land. The contractor shall be allowed, if he so desires to install hand pump(s)/tube well(s) at site of work at places as directed by the Engineer-in –Charge and nothing shall be charged from the contractor on this account. Contractor shall remove the hand pump(s)/tube well(s) as and when asked to do so by the Engineer-in-Charge. GE, in any case on completion of work and before issue of completion certificate shall ensure removal of hand pumps/tube wells unless GE desires that these hand pump(s) /tube well(s) be left in positions, and the contractor agrees to do so without claiming cost thereof from the department. No compensation what so ever shall be admissible to the contractor if he is required to remove the pump(s)/tube well(s) before completion of work. Use of water from such source(s) shall only be permitted if found after testing potable and fit for construction purposes as per IS-456-2000. The water from such source(s) shall be got tested by the contractor from the laboratory approved by the GE. The contractor shall submit the test results to the GE who shall after satisfying himself, permit the contractor to use water from such sources. Testing charges shall be borne by the contractor.

10.0 ELECTRICITY

Electricity will not be supplied by MES. The contractor shall make his own arrangements for provision and distribution of electricity through portable D.G sets of suitable capacities or by any other means for all the purposes in the contract.

11.0 DAMAGE TO EXISTING BUILDINGS, ROADS AND DRAINS ETC,

The contractor shall be responsible to reinstate and make good at his own expense, the damage if any caused by his workmen etc. to existing buildings, roads, drains, cables etc., while carrying out the work under this contract. Rectification, reinstatement, replacement, making good and touching up, etc, shall conform to the standard of workmanship and materials, originally used and the same shall be done to the entire satisfaction of Engineer-in-charge.

12.0 QUARRIES AND ROYALTY

Reference condition 14 of IAFW-2249 General Conditions of contracts. Delete this condition in toto and substitute as under: -

"Neither quarries are available nor shall the quarrying be permitted on the defence land".

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13.0 OCTROI AND OTHER DUTIES

The price quoted by the contractor shall be deemed to include all taxes/ cesses viz GST . The contractor shall not include any condition for fluctuation in price of labour and/or materials in his tender. Variation in prices of labour and materials during the progress of work consequent to any act of legislature is already covered in condition 63 of IAFW-2249 as modified by special condition here-in-after.

14 **LABOUR WELFARE CESS TAX**. Labour welfare cess shall be levied at the rate of 1% (one percent) on gross payment to contractor. Labour welfare cess deemed to be included in the lump sum/contract sum quoted by the tenderer.

15.0 MINIMUM WAGES PAYABLE TO LABOUR

15.1 Refer condition 58 of IAFW-2249. The contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt. of India/State Govt. whichever is higher.

16.0 The fair wage referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages as referred to above.

16.1 The contractor shall have no claim whatsoever if on account of any local regulation or otherwise he is required to pay wages in excess of minimum wages as described above during the execution of work.

17.0 PERMIT FOR PLYING GOODS TRANSPORT IN J&K AREA

Recommendations for issue of permits for plying goods transport, private carrier for carrying materials for work will be made by the GE 969 EWS, C/O 56 APO to competent transport authority of J&K Govt. However, the Ministry of Defence does not guarantee issue of such permits. GE 969 EWS, C/O 56 APO reserves the right to decide the routes, number of vehicles to be recommended for issue of permits, duration for which the permits is to be recommended and his decision shall be final, conclusive and binding with reference to the matter referred to in this condition.

18.0 CO-OPERATION WITH OTHER AGENCIES

The contractor shall permit free access and afford reasonable facilities to other Agencies or departmental workmen engaged by Govt. to carry out other work services (whether or not in connection with the work under the scope of this contract) on the site, simultaneously. The contractor shall not be allowed any extra payment on this account.

19.0 CRITICAL PATH METHOD IN LIEU OF TIME AND PROGRESS CHART

(a) The time and Progress Chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed network analysis and a time schedule. The critical path network will be drawn jointly by the GE and the contractor soon after acceptance of the tender. The time scheduling of the activities will be done by the Contractor so as to finish the work within the stipulated time. On completion of the time schedule a firm calendar date schedule will be prepared and submitted by the Contractor to the GE who will approve it after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site.

(b) During the currency of work the Contractor is expected to adhere to the time schedule and his adherence will be a part of contractor's performance under the Contract. During the execution of the work the Contractor is expected to participate in the reviews and updating of the net work undertaken by the GE. These reviews may be undertaken at the discretion of the GE either as a periodic appraisal measure or when the quantum of work ordered on the Contractor is substantially changed through deviation orders or amendments. Any revision of the schedule as a result of the review will be submitted by the contractor to the GE within a week who will approve it after due scrutiny. The Contractor will adhere to the revised schedule thereafter. In case of Contractor disagreeing with the revised Schedule, the same

SPECIAL CONDITIONS

will be referred to the Accepting Officer whose decision will be final, conclusive and binding. GE's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

(c) The Contractor shall mobilize and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

(d) No additional payment will be made to the Contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule even though the time schedule is approved by the Department.

20. Contractor shall mobilise and employ sufficient resources to achieve the detailed schedule within the broad framework of the accepted method of working and safety. No additional payment will be made to contractor for any multiple shift work or other incentive methods of achieving the target dates.

21.0 SITE FOR EXECUTION OF WORK.

Site for execution of work will be available as soon as the tender is accepted. In case, it is not possible for the department to make the entire site available at the commencement of work, the contractor will have to prepare the CPM net work as required in terms of Special Condition 21 to achieve the target. No claim whatsoever for not giving the entire site on commencement of work and for giving the site gradually will be tenable.

22.0 SECURITY OF CLASSIFIED DOCUMENTS

22.1 Tenderers special attention is drawn to condition 2A & 3 of General Conditions of Contracts(IAFW-2249) for compliance. The contractor shall not communicate any classified information regarding the work either to sub contractor(s) or to any other person(s) without prior written approval of the Engineer-in-Charge. The contractor shall also not make copies of the design and drawings and other documents furnished to him in respect of works and he shall return all documents to the department on completion of the work or earlier determination and/or termination of the contract.

22.2 The contractor shall attach with final bill the receipt from Engineer-in-Charge to the effect that the contractor has returned the classified documents as per condition 3 of General Conditions of Contracts(IAFW-2249) .

23. **RECORD OF CONSUMPTION OF CEMENT**

23.1 For the purpose of keeping a record of cement procured/received and consumed in work, the contractor shall maintain a properly bound register in the form approved by the Engineer-in-Charge showing the daily receipt of cement, quantity used in works and balance at the end of each day and such other information as is required to control the consumption and procured/receipt of cement. This register shall be signed daily by the contractor's representative in token of correctness of the entries made and it shall be signed in token of check by MES representative as follows:-

(i) Daily by JE in charge of supervision of works.

(ii) Once in a week by the Engineer-in-Charge.

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23.2 The register shall be kept at site of work in the safe custody of the contractor during progress of work and on demand he shall produce the same for verification of Inspecting Officer. The register shall be returned to MES on completion of work.

23.3 The aforesaid provisions will however not absolve the contractor to use required quantity of cement in work

24.0 CLEANING DOWN

The contractor shall clean all floors, joinery, glass panes etc. and remove drops of cement/lime paint etc. and touch up all painter's work and carry out other items of work as are necessary to make and leave the premises clean and tidy before handing over the buildings and/or works. The contractor shall not be allowed any extra payment on this account.

25.0 VALIDITY OF TENDER

The tender shall remain open for acceptance for a period of 60 days (Sixty days) from bid submission end date.

26.0 PROPRIETARY MATERIALS/ARTICLES AND RECORD OF MATERIALS

(a) The proprietary materials/articles such as paint, bitumen, water proofing admixture, etc. when brought to site shall be inspected and approved by the Engineer-in-Charge.

(b) The quantity of proprietary articles brought to site shall be recorded in measurement book (IAFW-2261) and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the works.

(c) The proprietary materials shall be stored as directed. Those already recorded in the measurement book shall be marked suitably for identification.

(d) The contractor shall produce original receipt vouchers/invoices including manufacturer's test certificates (where applicable) in respect of the supplies to the GE to ensure that the contractor has actually brought the required quality and quantity from the authorised source(s) and also to find out the rates thereof.

(e) Under no circumstances payment on account (RAR) shall be made to the contractor as material lying at site for incorporation in the work, unless the contractor produces original purchase vouchers/invoices for the following items of materials as applicable:-

(i) Water proofing admixture.

(ii) Paint, distemper, varnish, cement base paint and the like.

(iii) CI pipes & fittings (both water supply and soil/waste/ vent pipes) and valves.

(iv) Sanitary fittings.

(v) Factory made Steel windows and ventilators.

(vi) Factory made shutters

(vii) Floor/ wall ties

(viii) Iron mongery

(ix) SGSW pipes

(x) Electrical and water supply fittings/ fixtures where names of manufacturers/brands are specified or approved.

(xi) Cables.

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(xii) Cement

(xiii) Steel.

(xiv) Tarfelt /bituminous products.

(f) Payment for the above materials shall neither be made in the RARs nor these shall be allowed for incorporation in the work unless the purchase vouchers invoices as stated above are produced by the contractor and are verified by the Engineer-in-Charge regarding their correctness and genuineness and are defaced by him against the contract.

27.0 REFER SPECIAL CONDITIONS 6 & 9

(i) The contractor shall maintain site/area of land/buildings in a clean and tidy conditions at his own expenses and shall deliver up the same on/or before the date of completion of the works, or on termination of the contract, or in the event of the said areas or land/buildings required by the GE within one month of an order to that effect, in a clean and tidy state complete in every particular and to the satisfaction of Engineer-in-Charge.

(ii) In addition to completion of all other stipulations under condition 49 of IAFW-2249, the completion certificate under condition 49 of IAFW-2249 shall be issued only after vacation of site/area of land/building allotted to the contractor. However the contractor may be permitted by the GE in writing to retain such areas of land/building which is in the opinion of GE may be required for storage of materials required for rectification of defects (decision of the GE shall be final and binding in this regard). Soon after the rectification of defects such area of land/building shall be vacated and delivered to the Engineer-in-Charge in a neat state. Payment of final bill shall be made to the contractor only after such areas of land/building are vacated by the contractor.

28. **OFFICIAL SECRET ACT:** The contractor shall be bound by the Indian official Secret Act 1923 particularly Section 5 thereof, all as per condition 2A of IAFW-2249.

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29 SPECIAL CONDITION FOR CONCILATION

29.1 Scope of conciliation.

29.1.1 The scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated therein.

(a) Disputes relating to levy of compensation for delay in completion. actual amount of compensation.

(b) Disputes relating to technical examination of works.

(b) Disputes relating to interpretation of the provisions of the contract with reference to their application to parties.

(c) Disputes relating to non return of Schedule ‘B’ stores/over issued to contractor.

(d) Any other disputes having fair chances of being resolved by conciliation and considered fit to be referred to conclusion by the parties.

For item (b), (c), (d) & (e) each as stated above the financial limit shall be Rupees two lakhs or one percent of the contract amount which ever is less.

SPECIAL CONDITIONS

29.2 Commencement of conciliation proceedings.

29.2.1 The party initiating conciliation shall be send to the other party a written invitation to conciliate briefly identifying the subject of the dispute.

29.2.2 Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.

29.2.3 If the other party rejects the invitation there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply with in 30 days from the date on which he sends or within such other periods of time as specified in the invitation. he may elect to treat this as a rejection of the invitation to conciliate and if he so elects, he shall inform in writing the other party accordingly.

29.2.4 Number of conciliators .There shall be a Sole conciliator.

29.5 Appointment of conciliator : All disputes brought out in para 33.1.1(a) to (e) above shall be referred to the Sole Conciliator viz serving Officer not below the rank of superintending Engineer/ Superintending Engineer (QS&C) having degree in Engineering or equivalent or having passed final/Direct final examination of sub division-II of Institution of surveyors(India) to be appointed by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the Officer officiating as Engineer-in-Chief or Director General of Works specifically delegated by the Engineer-in-Chief in writing.

29.6 Status of Effect of Settlement Agreement

The settlement agreement signed by the parties as a result of conciliation proceedings shall be the same status and effect as it is an arbitral award on agreed terms.

30. QUALIFIED TRADESMEN (Applicable for estimated cost one crore and above work as per NIT)

In compliance with the condition 26 of IAFW-2249(General Conditions of Contracts) the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/National Institute of Construction management and Research (NICMAR)/Similar reputed and recognized Institutes by State/Central government to execute the works of their respective trade. The number of such qualified trades men shall not be less than 25% of total skilled/semi skilled tradesmen required in each trade. The Contractor shall submit the list of such tradesmen alongwith requisite certificates to Garrison Engineer for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades leading to un-satisfactory workmanship, the contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE. GE's decision whether particular tradesmen possess requisite qualification, skill and expertise commensurate with nature of work shall be final and binding. No compensation whatsoever on this account shall be admissible

31. RE-IMBURSEMENT/REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE" :-

(a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes/ cesses viz GST, duties, Royalties, Octroi & other levies payable under the respective Statutes. No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies, and/or imposition/abolition of any new/existing taxes / cesses, duties, Royalties, Octroi & other levies shall be made except as provided in sub para (b) here-in-below.

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(b) (i) The taxes / cesses which are levied by Govt at certain percentage rates of Contract Sum/Amount shall be termed as “taxes directly related to Contract value” such as GST Labour Welfare Cess/tax and like as applicable but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all “taxes directly related to Contract Value” with existing percentage rates as prevailing on last due date of receipt of tenders. Any increase in percentage rates of “taxes directly related to Contract value” with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the Contractor and any decease in percentage rates of “taxes directly related to Contract value” with reference to prevailing rates on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from any payments due to the Contractor. Similarly imposition of any new “taxes directly related to Contract value” after the last due date for receipt of tenders shall be reimbursed to the Contractor and abolition of any “taxes directly related to Contract value” prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from the payments due to the Contractor.

(ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further “taxes directly related to Contract value”, give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating there to which he may be in a position to supply. The Contractors shall submit the other documentary proof/informations as the GE may require.

(iii) The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other informations as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of “taxes directly related to Contract value” shall be made only if the Contractor necessarily & properly pays additional “taxes directly related to Contract value” to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt Authority and submits documentary proof for the same as the GE may require.

32. **SITE LABORATORY AND TESTING OF MATERIALS**

(a) **A LEVEL TESTS FOR WORKS COSTING Rs 100 LAKHS AND ABOVE:**- The contractor shall set up site laboratory of minimum size 3m x 4m with proper ventilation and lighting for testing of materials (except Schedule ‘B’ materials) for A level tests as listed in Appendix ‘F’ to Particular specifications. The contractor shall arrange all equipments/ machines for the tests specified in Appendix ‘F’ as A level tests at his own cost with prior approval of GE. The cost shall be included in the lump sum costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-Charge. The successful test results thereof shall be recorded and signed jointly by the contractor and Engineer-in-Charge. The charges of these tests i.e. A level tests carried out in site laboratory of the contractor shall not be recovered. However in case a test which was specified to be carried out in site laboratory and which could not be carried out in site laboratory due to some reason, such test shall be carried out in Zonal lab/ any other approved lab and the recovery will be made at the rates given in Appendix 'F' for tests carried out in Zonal lab or other wise at the actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the tests results for such tests.

SPECIAL CONDITIONS

(b) **A LEVEL TESTS FOR WORKS COSTING UP TO Rs 100 LAKHS:-** The contractor may set up site laboratory at his option for works costing up to 100 lacks. The other stipulations will be same as specified in preceding Para (a). However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved/ set up but GE, the recovery shall be made at the rates given in Appendix 'F' for tests carried out in Zonal lab or otherwise as per actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the test results for such tests.

(c) **B & C LEVEL TESTS**: For tests of 'B' and 'C' level as indicated in Appendix 'F', the contractor shall provide all facilities for testing of materials at Zonal Laboratory/ Govt approved laboratories or test houses/ Engineering Colleges at his own cost. The lump sum/ rates quoted by the contractor shall deem to be inclusive of these tests. For 'C' level tests, actual cost of testing shall be borne by the Contractor and for 'B' level tests recovery shall be made at the rates of various tests conducted in Zonal Laboratory of MES as indicated in Appendix 'F'. Wherever it is convenient to get 'B' level test done at approved test house/ Engineering College, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same.

(d) The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective test or whenever the test is due whichever is earlier.

(e) This clause supersedes all other stipulations, if any, with regard to testing of materials specified elsewhere in the tender documents.

33. **Testing of Materials**

(a) The cost of testing of materials as listed in Annexure 'I' to special conditions except materials issued under Schedule 'B' shall be borne by the contractor.

(b) Tests of various materials as specified in SSR Part I shall be carried out.

(c) Rates given under Col 7 of Annexure 'I' shall be per lot of samples.

34. **EMPLOYEES PROVIDENT FUND SCHEME** :-

(a) Under the Employees Provident Funds and Miscellaneous Provisions Act, 1952, the contractor shall have a EPF Code No and all labours employed by him or sub-contractor shall be enrolled as member of Provident Fund and should be given Universal Account Number (UAN).

(b) A certificate to effect that all working labours employed for execution of work are registered for EPF and due distribution have been credited into their account, shall be furnished by contractor along with each running advances or final bill.

(c)The contractor shall not pay wages lower than the minimum wages for labour fixed by the Govt of India / state Govt /Union territory whichever is higher. The contractor shall submit the proof of crediting the wages in the Aadhar linked bank account of the contractor which shall be prerequisite to release subsequent payment to the contractor.

35. Blank

38. Performance Security.

(i) Within 28 days of receipt of the Letter of Acceptance, the successful contractor in shall deliver to the Accepting officer a performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum.

(aa) A Bank Guarantee in the prescribed form.

(bb) Government Securities, FDR or any other Government instruments stipulated by Accepting officer.

(ii) If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

(iii) The period of validity of bank guarantee bond against performance security shall be upto and including Sch of expiry of defect liability period.

(iv) Failure of the successful contractor to comply with the requirements of sub clause 38 (i) above shall constitute grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor. Amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

39. REFUND OF PERFORMANCE SECURITY:-The performance security deposit as mentioned as above (as amended vide amendment No 45 to condition 68 of IAFW-2249) shall be refunded to the contractor after the expiration of the defect liability period provide always that the contractor shall first have been paid final bill and have rendered a No-Demand Certificate (I.A.F.W-451). Recovery/withholding due amount if outstanding against the contractor shall be effected from the Performance Security as stipulated under condition 67 of IAFW 2249 (as amended vide amendment No 44)

40. DEFECT LIABILITY PERIOD:-The defect liability period shall be TWENTY FOUR CALENDAR MONTHS after the work has been handed over to Government i.e. after actual date of completion (as per condition 46 of IAFW 2249 amended vide amendment No 48)

Signature of the Contractor

AGE(Contracts)
For Accepting Office

PARTICULAR SPECIFICATIONS

1. GENERAL

(a) Work under this contract shall be carried out in accordance with Sch 'A', particular specification and general specifications and other provisions contained in MES Standard Schedule of rates Part I of **2009** and part-II of **2020** here-in-after called as 'MES Schedule' and all these documents shall be read in conjunction with each other.

(b) Term "General specification" referred to here-in-before as well as referred to in IAFW-2249 (General conditions of contracts) shall mean the specifications contained in the MES Schedule.

(c) General rules, specifications, special conditions and all preambles in the MES schedule shall be deemed to be applicable to the work under this contract unless specifically stated otherwise in Schedule 'A' and/or in particular specifications, in which case the provisions contained in these documents shall take precedence over the provisions contained in MES schedule. The term "as specified" wherever appears in tender documents and drawings relates to relevant particular specifications and in its absence general specifications.

(d) Where specifications for any item of work are not given in these particular specifications or in MES Schedule the specifications as given in relevant Indian Standard or code of practice shall be followed.

1.2 Reference to some paragraphs of MES Schedule have been made in these particular specifications but other paragraphs and provisions as applicable shall also be followed.

1.3 MATERIALS

All the materials to be procured and incorporated by the contractor in the all work under this contract.(with the exception of local materials like, bricks, stone aggregate, stones and sand etc) shall bear IS certification mark unless the materials required for incorporation in this work is /are not manufactured with IS certification mark in the country or where materials of specific Make (s) is/are indicated either in Schedule 'A' or in particular specification hereinafter and the same is /are not manufactured with ISI certification mark. A list of manufacturers holding valid licenses to use ISI certificate mark published by ISI Manak Bhawan, Bahadur Shah Zafar Marg, New Delhi-110002 shall be referred for the materials manufactured with ISI certification mark.

1.3.1 Such materials which are not manufactured with ISI certification mark shall comply with the requirements of relevant ISI specifications and conform to samples displayed in CWE's / GE's office

1.3.2 Indian Standard (IS) of the year of publication/edition listed or specified in the SSR (Part-I) shall be applicable for the work under this contract unless specifically indicated otherwise elsewhere in these tender documents. The contractor may however at his own discretion opt for corresponding subsequent/amended or revised editions of the IS, without any adjustment to contracts sum.

1.3.3 Local materials such as sand, bricks, aggregate etc shall conform to the specifications and conform to the samples kept in the office of CWE's/GE's and the tenderers are advised to inspect these samples. The materials to be provided by the contractor shall conform to or shall be superior in quality than these samples, the tenderers shall be deemed to have inspected these samples before quoting and have full knowledge thereof whether they inspect them or not.

1.3.4 PROPRIETARY MATERIALS

1.3.4.1 Proprietary materials shall be procured and brought to site in manufacturers original sealed containers/packing with seal and marking on the container/packing intact except where the requirement for the whole work is less than the contents of smallest container/packing.

1.4.1 **SAMPLE OF MATERIALS: -**

(a) Refer condition 10 of IAFW-2249

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(b) Specific requirements, regarding dimensions strength weight and finishes as per IS, SSR and the particular specifications given here-in-after vis- a- vis actual properties check, tests carried out, reference to test certificates and marking, etc based on which samples of each materials are approved as conforming to relevant specifications shall be recorded in the sample approval register.

(c) The contractor shall not procure materials unless the samples are first got approved from the Garrison Engineer/EIC.

1.4.2 **RECORD OF MATERIALS:-**

(a) The quantity of materials such as paints, Liquid water proofing admixture and the like which looses their identity after incorporation in the work as directed by the Engineer-in-Charge, shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that their required quantity has been brought at site for incorporation in the work.

(b) Materials brought to site shall be stored as directed by the Engineer-in-charge and shall be suitably marked for identification.

(c) The contractor shall on demand produce to the GE/EIC, original purchase vouchers / invoices in respect of the supplies. Vouchers/ invoices so produced and verified shall be stamped by the Engineer-in-Charge indicating contract number. The contractor shall ensure that the materials are brought to site, in manufacturer's original sealed containers/packing bearing manufacturer's marking except in the case of requirement of materials being less than smallest packing.

(d) The contractor shall produce to the GE in original printed and machine numbered purchase vouchers/ invoices including manufacturers test certificate for the materials shown here in after. (as applicable and as required for incorporation in the work), whenever procured and brought to site of work for incorporation in the work. A Xerox copy of defaced original Vouchers/ invoices by the Engineer-in-Charge there of stating "verified for materials purchased and brought to the site of work for incorporation in the work under CA" and signed with date shall be submitted to the GE/EIC.

(i) Steel
(ii) Cement
(iii) All electrical & mechanical equipment s
(iv) All such material for which the payment as "Material lying at site" is being claimed in the RARs.

Payment for above materials shall neither be made in the RARs nor these will be allowed for incorporation in the work unless purchase vouchers/ invoices and test certificate of manufacturer as stated above are produced by the contractor and are verified by the Engineer-in-Charge.

1.4.3 The quantity of all the proprietary materials (including the materials, the quantity of which can not be checked after incorporation in the works as stipulated in clause 1.6.2 (a) here in before) shall be recorded in Measurement Books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought at site for incorporation in the work.

1.5 STANDARD OF WORKMANSHIP

To determine the acceptable standard of workmanship and layout of fittings, wiring etc GE may order the contractor to execute certain typical portion of work (different trades) and services sufficiently in advance of other work if so desired by the Accepting Officer. These shall be executed and completed under the close supervision of the Engineer-in-Charge and shall be got approved from GE. On the approval by the Accepting Officer of such items, these items shall be signed and/or suitably identified by the Accepting Officer and labelled as guiding samples. Approved finishes/workmanship shall be followed in the work as a whole.(not applicable for Term Contracts)

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1.6 **SCOPE OF WORK**

The scope of work under this contract comprise of entire completion of work as per Schedule 'A' and in General Summary (Where applicable) all as specified in these particular specifications and General Specifications .

2. DEMOLITION AND DISMANTLING

2.1. Refer section 21 of SSR- Part –I in General

2.2 The tenderers shall visit the site of work to assess nature and extent of work involved in demolition/dismantling and the quantity and quality of material obtainable there from. The tenderer shall be deemed to have visited the site and availed the pre-tender opportunity to assess all the eventualities, which may affect his quotation whether he actually visits or not.

2.3 The contractor shall take all necessary precautions for the protection of adjoining property/building while taking up work of dismantling/demolition. Any damage to exiting property/building shall be made good by the contractor without any extra cost to the department.

2.4 The material obtained under Sch of credit shall be handed over to the contractor by the EIC on a proper gate pass and credit for demolished/dismantled material for these items will be deducted from the RAR paid subsequent to demolition/dismantling of the work catered for is executed. But in no case the material shall be allowed to be removed from the site until the credit has been recovered by the Govt. The materials beyond credit schedule, if any, retrieved shall be the property of the Govt & shall be deposited by the contractor in the MES store yard without any extra payment

2.5 The contractor will remove demolished/dismantled material serviceable and unserviceable material from the site of work without any extra cost to Govt.

2.6 **Safety in construction**: Refer clauses 1.16 and 21.6 of MES Standard Schedule of Rates Part-I (2009

3. **EXCAVATION AND EARTH WORK**

3.1 Excavation General

(a) Excavation and earthwork under this contract shall be considered as excavation in soft/loose soil as detailed in MES Schedule.

(b) Unit rates for items in Schedule 'A' shall include for excavation and earthwork in soft/loose soil. No extra payment shall be admissible to the contractor in case the strata met with actually at site is found different from that mentioned hereinbefore. Any deviation involving excavation and earthwork will be priced on the basis of soft/loose soil.

(c) Boulders and stones obtained from excavation shall be sorted out and neatly stacked as directed by Engineer-in-Charge and shall be removed to MES store yard without extra cost to the Govt.

(d) In case timbering to excavation is required and specifically ordered by the GE in writing, it shall be paid as deviation.

(e) Bailing/pumping out of water where required shall be carried out as described in clause 3.17 of MES SchedulePart-II. The cost of such work as may be necessary shall be deemed to be included in the unit rates quoted by contractor. No additional payment as stipulated in para 3.17 of MES Schedule Part-II shall be admissible. In the event of deviation no price adjustment shall be made for cost of bailing/pumping out.

(f) All rubbish obtained form site clearance and all spoil obtained from surface-dressing shall be removed, spread and leveled as directed by Engineer-in-Charge.

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3.2 **FILLING IN TRENCHES, PLINTH AND UNDER FLOORS ETC**.

3.2.1 Filling excavated earth in foundation trench shall be carried out as per paras 3.19.1 to 3.19.3 of MES Schedule Part I. Earth obtained from excavation and not approved for filling or surplus to the requirement shall be removed to a distance as specified in schedule 'A' or as directed by the Engineer-in-Charge.

3.2.2 All filling shall be spread in layers not exceeding 25cm thick (each layer) watered and well rammed. Thickness of filling as mentioned in Schedule 'A' or indicated also shall be after consolidation.

3.2.3 For workmanship and other specifications refer paras 3.22,3.22.1 to 3.22.12 of MES Schedule Part-I. The filled earth shall be spread in layers, rolled and compacted to achieve 95% of the standard proctor density for each layer.

3.2.4 **Hard core** : Hard core shall be of brick (broken) of gauge not exceeding 63mm. Hard core shall be deposited, spread and leveled in layers n.exc 15cm thick and watered and well rammed to a true surface and compacted. The thickness of the hard core specified or as indicated in the schedule 'A' or shown on drawings, is the thickness after consolidation and where not specifically mentioned, it shall be 150mm consolidated thickness.

4. **CONCRETE**

4.1 CEMENT

4.1.1 Type of Cement: The type of cement to be selected for use with invariably depend upon the specific usage in the work(s). 33 - grade OPC will not be used in Works. The use of PPC of required strength with the fly ash content as per IS 1489-1991 Part-I is also permitted. The details of various types and grade of cement are enumerated below :-

(i) Ordinary Portland cement, grade 43, IS-8112-1989&grade 53, IS-12269 -1987.
(ii) Rapid Hardening Portland Cement (IS-8041-1990)
(iii) High Alumina Cement (IS-6452-1989)
(iv) Sulphate Resisting Portland Cement (IS-12230-1988)
(v) Portland Pozzolana Cement, grade 43, IS-1489-1991 (Part-I), subject to certain conditions given in clause 3.1.3 hereinafter. These may be read in conjunction with the provisions contained in SSR Part-I 2009,Section 4(Para 4.3).

4.1.2 **PROCUREMENT OF CEMENT** :Cement shall be procured from the main producers of the cement as specified in Appendix 'B-1'

4.1.2.1 Where estimated requirements of cement is less than **1200** bags, the contractor can procure cement from the authorized distributors/dealer of the approved firms. However, contractor will have to submit test certificate of the batch issued by the main producers.

In the above case, the independent testing will be at the option/discretion of the Garrison Engineer, if the factory test certificate is produced by the contractors. In a particular case when the contractor does not produce the factory test certificate, the testing of cement by Garrison Engineer will be mandatory. However, the testing will be restricted to IS-4031 (Methods of physical test for hydraulic cement only).

4.1.2.1 The Particulars of the manufacturer of cement alongwith the date of manufacture shall be produced by the contractor for every lot of cement separately. The cement so brought shall be fresh and in no case older than 60 days from the date of manufacturing. The documents in support of the purchases of cement such as purchase voucher shall be verified by the GE. Before placing the order for supply of cement by the contractor, he shall obtain written approval from the GE regarding name of manufacturer, quantity of cement etc. Cement shall be procured for minimum requirement of one month and not exceeding the requirement of the same for more than TWO months at a time. The cement shall be consumed in the work within THREE months after receipt. Cement shall conform to the requirement of Indian Standard Specification and each bag of cement shall bear relevant ISI mark. The weight of each consignment shall be verified by the GE and recorded.

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4.1.2.3 The cement shall be procured by the contractor preferably in paper polyethylene bags. The Average weight of each bags cement shall not be less than 50 Kg. The average weight shall be tested by taking the weight of 50 Kg bags selected at random by GE per lot. In case the weight is less than 50 Kg whole lot shall be rejected. In case weight is more than 50 Kg it shall be deemed to contain 50 Kg only and nothing extra shall be admissible to contractor on account of extra cement used in the work.

4.1.2.4 Cement shall be procured by the contractor directly from the following producers:

(a)	The Associated Cement Co. Ltd (ACC)	
(b)	Grasim Industries Ltd (GRASIM)	
(c)	Ultra Tech Cement (ULTRATECH)	
(d)	The India Cement-	
(e)	Dalmia Cement (DALMIA)	
(f)	Andhra Cement Ltd	
(g)	Century Cement (CENTURY)	
(h)	Saurashtra Cement (SAURASHTRA)	
(j)	Binani Cement (BINANI)	
(k)	M/s Heldelberg Cement India Ltd (MYCEM), Gurgaon- 122002 (MYCEM)	
(l)	Mangalam Cements (MANGALAM)	
(m)	Birla Corp Ltd (BIRLA)	
(n)	Orient Cement (ORIENT)	
(o)	Lafarge Cement (LAFARGE)	
(p)	Shree Cement (SHREE)	
(q)	JK Cement (JK) Kanpur	
(r)	Jaypee Rewa Cement (JAYPEE)	
(s)	Ambuja Cement (AMBUJA)	
(t)	JK Lakshmi Cement (JK LAKSHMI)	
(u)	M/s The Ramco Cements Ltd, Auras copraporate centre , 98-A Chennai -600004 (RAMCO)	
(v)	Blank	

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(w)	Dhuruv Industrial Company Ltd (DHRUV)	
(x)	Shree Guru Kripa Cement (Pvt) Ltd (SARTAJ)	
(y)	M/s Prism Cement Ltd, 3/113, Vivek Khand, Gomti Nagar, Lucknow -226010 (PRISM)	
(z)	M/s Zuari Cement Ltd No.6 MC Nichols Road Chetpet, Chennai-600031 (ZUARI)	
(aa)	M/S Toshali Cements Pvt Ltd 9-14-13 C.B.M Compound Near Mehar Apartments Visakhapatnam-530003 (GAJAPATI)	
(ab)	M/s Cement Manufacturing Company Ltd, Mayur Garden, 2nd Floor, GS Road, Guwhati-781005 (STAR)	
(ac)	M/s OCL Inida Ltd, Rajgangpur, Distt- Sundargarh, Orissa-770017 (KONARK)	
(ad)	M/s Chettinad Cement Corporation Limited, 4th Floor Rani Seethai Hall Bldg Chennai- 600006 (CHETTINAD)	
(ae)	M/s My Home Industries Ltd, Mahapur, Hyderabad-500081 (MAHA)	
(af)	M/s Parasakti Cements Ltd, Srinivasa Nagar Colony (West) Hyderabad-500038 (PARSAKTI)	
(ag)	M/s Barak Valley Cements Ltd, Karimganj, Assam-784803 (BVCL)	
(ah)	M/s Sanghi Industries Ltd, Makarba, Ahmedabad- 380051 (SANGHI)	

4.1.2.5 Contractor shall produce purchase vouchers bearing machine number, test certificates and other valid documentary proof to the entire satisfaction of GE for entire quantity of each consignment to prove that cement has been purchased from the main producer or their authorized dealers/stockiest as listed in Para 4.1.2.4 above. The contractor shall also submit particulars of the manufacturer of cement for every lot for verification by the site staff and GE.

4.1.3 GUIDE LINES FOR **USE OF PORTLAND POZZOLANA CEMENT (PPC)** :While using PPC, the following requirement generally be met :-

(i) PPC shall meet the strength criteria of 43 grade Ordinary Portland cement (OPC) as laid down in IS-8112-1989.

(ii) Minimum period before striping the form work given in clause 11.3.1 of IS 456 : 2000 and MES Schedule Part-I (SSR Part-I) vide clause 4.11.6.3 shall be increased by seven days. For example the minimum period before striking form work for soffits of beams (props left under) given as 7 days in clause 4.11.6.3 (c) of SSR Part-I shall be 14 days (7+7) for PPC.

(ii) Both OPC & PPC shall not be permitted for use in the same building except for plaster and mortar.

(iii) While procuring PPC, the mandatory certificate of testing and quality assurance will continue to be submitted as hither-to-fore, with fly ash content strictly as per IS 1489 Part-I.

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USE OF FLY ASH

(a) Flyash is interground with clinker and not mixed with clinker.
(b) Dry flyash is transported in closed containers and stored in celcius. Only pneumatic pumping shall be used.
(c) The flyash shall be from thermal plants using high temperature combustion above 1000 degree celcius.
(e) The flyash content in PPC shall not exceed 25% to ensure consistency.

(iv) In cold climatic conditions where temperature goes below 15 degree centigrade and Important structure like overhead reservoir, underground sumps, prestigious building (as decided by CWE) and buildings with span 10m or more, only OPC shall be used. However, for other structure during working seasons in cold climatic regions, where OPC is not available, use of PPC may be permitted provided it shall be ensured that while using PPC, atmospheric temperature should be more than 15 degree centigrade.

4.1.4 TESTING OF CEMENT

4.1.4.1 The manufacturer is to carryout inspection and testing of cement in accordance with relevant BIS provisions. The contractor shall submit the manufacturer's test certificate in original alongwith test sheets giving the results of each physical test as applicable in accordance with relevant IS provision and the chemical composition of cement or authenticated copy thereof duly signed by the manufacturer with each consignment, as per the following IS provisions :-

(a) Method of sampling for hydraulic cement as per IS-3535-1986.
(b) Method of physical test for hydraulic cements as per IS-4031.
(c) Method of chemical analysis of hydraulic cement as per IS-4032-1985.

4.1.4.2 The test certificate and test sheet shall be furnished with each batch of manufacture. The Engineer-in-Charge shall record these details in cement supply acceptance register to be maintained by him which will be signed by Supdt B/R Gde-I/JE (Civil). Engineer-in-Charge, Garrison Engineer and Contractor as per Appx 'E' here-in-after.

4.1.4.3 The contractor shall however, organize setting time , fineness, soundness and compressive strength test of cement through designated laboratory on samples collected from the lot brought at site before incorporation in work. The contractor will be allowed to use the cement only after satisfactory compressive strength of seven days. To meet this requirement contractor is required to keep minimum 10 days stock before any new lot is brought at site which can be used in the work. The contractor shall be required to remove the cement not meeting the requirement from site within 24 hours. Seven day strength test will be rallied upon to accept the lot of cement to commence the work, 28 days compressive strength test will be the final criteria to accept/reject the lot.

4.1.4.4 The GE shall carryout independent testing as per the tests mentioned in the 'CEMENT SUPPLY/ACCEPTANCE FORM' of random samples of cement drawn from various lots, the random sample of cement to be tested shall be drawn as per **quality assurance manual**.The GE shall also organize independent physical test from NABL Laboratories like National Test house /SEMT wing CME Pune/Regional Research Laboratories/IIT's, MES Zonal Laboratories, Govt Engg Colleges and National Institute of Technology as per relevant BIS and shall be recorded in the respective portion of Appendix 'B'. The decision as to where the testing of cement is to be done shall be taken by GE. In case the cement is not of requisite standard despite manufacturer's test certificate, the contractor shall remove the total consignment from the site at his own cost after written rejection order of the consignment by the GE.

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4.1.4.5 The random samples as per relevant IS shall be selected by GE before carrying out testing. The record of such samples selected by the GE for testing shall be properly maintained in the 'Cement Testing Register' giving cross reference to relevant consignment of cement and quantity received etc.

4.1.4.6 Cost of transportation of samples to the approved laboratory/test house and all testing charges including cost of sample shall be borne by the contractor.

4.1.4.7 Notwithstanding the submission of purchase vouchers/test certificate, following tests shall be mandatory carried out as per IS-4031 :-

(a) Initial and final setting time.
(b) Soundness test.
(c) Compressive strength test at 3 days, 7 days & 28 days as specified in the relevant Paras of the Indian Standard.
(d) Consistency test.
(e) Fineness.

Other tests need not be carried out for cement procured with BIS certificate mark unless specifically asked by GE or Accepting Office and the contractor shall not have any claim on this account.

4.1.4.8 The cement shall conform to chemical requirements and physical requirements as specified in the respective clauses of the Indian Standard. The tests carried out as per provisions of IS codes specified here-in-before, shall be criteria for acceptance of cement by Engineer-in-Charge. If samples from a lot/lots are not within the acceptable limits of Indian Standard, the lot/lots shall be rejected without any claims of compensation to the contractor for the lot/lots purchased. The contractor shall replace the lot/lots with the fresh one which shall be tested again for acceptance.

4.1.4.9 The cost of all tests carried out on cement before acceptance for incorporation in the work shall be borne by the contractor whether the results are acceptable or not. No claim or compensation shall be entertained on this account and clause of IAFW-2249 deemed to be amended accordingly. Any cement lot/lots on being tested found not to be conforming with the relevant clauses of Indian Standard shall not be paid as material at site.

4.1.5.0 <u>STORAGE OF CEMENT</u>

4.1.5.1 The cement shall be stored in such a manner as to prevent deterioration due to moisture or intrusion of foreign matters. Refer clause 4.3.1 of SSR Part-I 2009 for storage and precaution in storage of cement. The height of platform shall be at least 20 cm from floor. Joint inspection of storage of cement shall be carried out by Engineer-in-Charge and contractor once in a day. The stacking of cement shall not be more than 10 bags high. The cement fully set or partially set shall not be used in the work and shall be removed by the Contractor immediately without any extra cost.

4.1.5.2 Tested and untested cement shall be segregated and stored separately with distinct identification. The cement godown shall be provided with two lock on each door. The key of one lock of each door shall remain with the Engineer-in-Charge or his representative and that of other lock with the contractor or his authorized agent at site of work so that cement is removed from the godown only according to the daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag(s) found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written order of the GE and suitable replacement for the cement bag(s) as removed shall be made and no claim whatsoever shall be admissible on this account.

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4.1.5.3 DOCUMENTATION

4.1.5.4 The contractor shall submit original vouchers form the manufacturer for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignment received at the work site shall be inspected by the GE alongwith the relevant documents before acceptance. The original Vouchers and the Test Certificate shall be defaced by the Engineer-in-Charge and kept on record in the office of GE duly authenticated and with cross reference to the consignment number/control number recorded in the cement acceptance register clearly mentioning 'Batch No' and 'Date of manufacture'. The cement acceptance register will be signed by JE(Civil), Engineer-in-Charge, GE and the contractor. The Accepting Officer may order BOO for random check of cement and verification of connected documents.

4.1.5.5 The entire quantity of all cement shall suitably be recorded in the measurement Book (as not to be abstracted) for record purpose before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

4.1.5.6 In addition to the cement Acceptance Register, the cement consumption register shall be maintained as stipulated in special conditions here-in-before.

4.1.7.0 **SCHEDULE AND SUPPLY**

4.1.7.1 Schedule of supply of cement shall be finalized by contractor with the GE and shall be incorporated in CPM chart so that supply of cement is monitored in a way to avoid any delay in completion of the work. The supply shall be regulated in such a way that cement is consumed in work three months of its procurement.

4.1.7.2 The complete requirement of cement will be worked out before making any RAR payment. Procurement of cement by the contractor shall be complete sufficiently in advance of the date of completion. The contractor will forfeit his right to demand extension of time if the supply of cement got delayed due to his failure in placing order in time to the manufacturer.

4.1.8 **GENERAL** : The payment shall only be allowed after production of original purchase vouchers, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (7 days compressive test) are found satisfactory after testing as specified here-in-before. Payment for cement shall be made as material laying at site as per condition 64 of IAFW-2249. Rate of cement given in SSR shall be applicable for cement irrespective of type and grade of cement specified for use in the work.

4.1.8.2 **COARSE AGGREGATE** : Coarse aggregate for all cement concrete works shall be broken/crushed stone aggregate. Aggregate shall be as specified in clause 4. 4. 1, 4. 4.2, 4.4.3.1, 4. 4. 3. 2 and 4. 4. 7 (1) of MES Schedule part-I.

4.1.8.3 FINE AGGREGATE : Fine aggregate for all cement concrete works shall be coarse sand and natural sand as specified in clause 4.4.1, 4.4.2, 4.4.3.2, 4.4.7 (2) of MES Schedule part-I as laid down for grading Zone- I & II.

4.1.8.4 WATER : Refer para 4.9 of MES Schedule part-I & stipulated in IS-456-2000.

4.1.8.5 MIXING AND CONSOLIDATION OF CONCRETE : Mixing, laying and consolidation etc of cement concrete in various situations shall be carried out all as specified in paras 4.11.1, 4.11.2, 4.11.2.1 to 4.11.2.4 of the MES Schedule part-I.

4.1.8.6 Finishing to exposed surfaces of concrete: Refer para 4.11.16 of MES Schedule Part-I.

4.1.8.7 FORM WORK: Refer para 4.11.6.1 to 4.11.6.5 and 7.15.1 to 7.15.10 of MES Schedule Part-I.

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4.1.8.8 REINFORCED CONCRETE : Against Schedule 'A', irrespective of whatever shown in drawings and irrespective of whether indicated in drawing or not, the RCC work shall be M-25(Design Mix). If in any drawing M-15 or M-20 has been shown for RCC work, the same shall be amended to M-25 (Design Mix) with reinforcement and size of members remaining the same. (For reinforcement refer clause 8.2 here-in-after). For other sections of Schedule 'A' the mix of RCC shall be as indicated against respective items.However for overhead reservoir (including staging and foundation) design mix M-30 concrete shall be used irrespective of whatsoever shown on drawings.

4.1.8.9 IMPORTANT REQUIREMENT OF REINFORCED CEMENT CONCRETE AND PLAIN CEMENT CONCRETE: The contractor shall provide all facilities such as material and labour, tools/equipment for moulding casting and conveyance of test cubes to laboratory situated at HQ CWE in case the facility is not available in HQ CWE lab for testing, contractor shall provide all facilities for testing in other department laboratories/Govt approved laboratories. Charges of testing shall be borne by the contractor and his quoted rates shall be deemed to include for the same.

4.1.9 **DESIGN MIX**

4.1.9.1 General : Design mix cement concrete shall be as per IS-456-2000. The provisions of this IS are further amplified as described here-in-after.

4.1.9.2 Design mix of concrete shall be got done by the contractor as per IS-10262 from laboratory/Engineering Colleges as approved by GE with available cement and aggregate to provide the grade of concrete at desired workability. The mix design calculation for trial and approved design mix shall be kept on record for future record. The expenditure incurred for testing shall be born by the contractor.

4.1.9.3 The GE shall be the final authority for accepting the mix design.

4.1.9.4 The quantity of cement arrived at the mix design shall be used in the work. However if the mix design gives lower value of cement contents, the minimum cement contents to be used in the concrete shall not be less than the quantities given as here-in-after :-

Type of Design Mix	Quantity of cement required per cubic metre of concrete
M-20 (Design mix)	320 Kgs
M-25 (Design mix)	385 Kgs
M-30 (Design mix)	420 Kgs

The other requirement shall be as per values given in IS-456 of 2000 amended upto date.

4.1.9.5 Irrespective of whatever shown in drawings and irrespective of whether indicated in drawing or not, the RCC work shall be M-25 (Design Mix). If in any drawing M-15 has been shown against RCC work, the same shall be amended to M-25 (Design Mix) with reinforcement and size of members remaining same. (For reinforcement refer clause 8.2 here-in-after). Irrespective of whatever quantity of cement is arrived at by Mix design, the cement contents shall be as per table 5&6 of IS-456:2000,and nothing extra shall be payable to the contractor on this account. The other requirement shall be as per values given in IS-456 of 2000.

4.1.9.6 As soon as possible after receiving the order to commence the work, the contractor shall submit the mix design from GE CTL(Command Test Laboratory) for the grade as specified as per Appendix 'C' of IS-10262 keeping into consideration provision of Para 3.6.1 to 3.6.4 here in above for perusal & approval by GE. Based on the mix design as above, contractor shall arrange to carry out trial mixes from GE CTL(to be approved by GE) using materials(cement, sand ,stone aggregates and water) from the site of construction as per Appendix 'D' of IS-20262.

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4.1.9.7 Details/results of trial mixes from the engineering establishment along with their recommendations regarding quantities of cement, water, sand and coarse aggregates to be used in the work to achieve characteristic compressive strength of 25 N/mm2 for M-25 design and 30N/mm2 for M-30 design Mix on field /site after 28 days shall be submitted by the contractor to the GE for final approval.

4.1.9.8 The approval of the GE shall not absolve the contractor of his responsibilities to obtain the required minimum strength of concrete at any stage.

4.1.9.9 Contractor's unit rates against respective parts of Sch 'A' shall include for the above provn and as brought out here-in-after. Further ,in case here is any change in source of local materials viz sand/aggregates or Make/grade of cement etc, the mix design shall be carried out and procedure here-in-before shall be repeated.

4.2 Batching : In proportioning concrete the quantity of both cement and aggregate shall be determined by mass and all in accordance with clause 10.2 and sub clauses thereon of IS 456-2000 as applicable.

4.2.1 Mixing : Mixing of concrete shall be done in a mechanical mixer and as per IS 456-2000 weighing by weigh batcher.

4.2.2 Trial Mixes : The actual mix proportion will be arrived at by means of number of trial mixes by changing the water cement ratio, proportions of fine and coarse aggregate/fineness module of aggregate by changing their grading and proportions etc. Attempt shall be made to make the mix design as economical as possible.

4.2.3 Sampling : The sampling procedure and the frequency of sampling shall be as per clause 15 of IS 456-2000.

4.2.4 Test Specimens : All test specimens shall be 150 mm cubes. For each sample, six cubes shall be cast out of which, three cubes shall be tested for 7 days and remaining three for 28 days compressive strength. The specimens shall be tested as described in IS 516-1959.

4.2.5 Test strength of sample : Refer clause 15.4 of IS 456-2000.

4.2.6 Accepting criteria : The acceptance criteria of the test result shall be as had down in IS 456-2000.

4.2.7 Workability : The workability of the concrete for different location shall be confirming To para 7.1 of IS-456-2000.

4.2.8 Compaction of concrete : Refer to Para 4.11.11 of MES Schedule Part-I consolidation compaction of RCC Slabs/beams/columns shall be done with approved mechanical vibrators. However where the same is not feasible compaction shall be achieved manually as directed by the Engineer-In-Charge

4.2.9 Packing and transportation of samples : Refer clause 13 of IS 456-2000. The contractor must at his own expenses properly pack the samples. The contractor shall bear the cost of transportation of the samples required to be tested from site of work to the laboratory where it is to be tested.

4.2.9.1 Variation in quantities of cement :

(a) The minimum cement content specified here-in-before shall be provided even if the laboratory mix design gives lower value of cement content.

(b) As a result of the mix design, if the quantity of cement required in order to obtain the desired properties of concrete varies beyond 5% of the minimum quantity of cement specified, the mix design shall be got approved from the Accepting Officer before incorporation in the work.

PARTICULAR SPECIFICATIONS(CONTD……)

(c) If the laboratory results of **Design Mix** works gives higher values of cement content than the minimum quantity specified hereinbefore, same shall be provided by the contractor and no financial adjustment shall be made for the same, No claim whatsoever shall be entertained on this account.

(d) In case of dispute with regard to provisions mentioned above, the decision of Accepting Officer shall be conclusive, final and binding on contractor.

4.2.9.2 The following documents shall be maintained at site:-

(a) Test reports and manufacturers certificate for material.
(b) Concrete mix design details.
(c) Pour cards for site organization and clearance for concrete placement.
(e) Record of site inspection of workmanship and field test.
(f) Non conformance reports, change orders.
(g) Quality control charts and statistical analysis.
(h) Statistical analysis.

4.2.9.3 RCC other than design mix shall be all as per relevant IS code and SSR.

4.2.9.4 Precast Reinforced concrete for Fencing post:-Refer para4.22 of MES schedule part-I

5. BORING HOLES (DRILLING OF BORE AND COLLECTION OF SAMPLES)

5.1 The drilling shall be carried out by approved method so that samples of bore holes shall be obtained in crushed form and not as come as obtained in clay drilling. Bore Hole of 450mm dia suitable for 200mm dia tube well shall be drilled upto 60m or 90m as ordered by GE below ground level or upto the bottom of basement/metamorphic rock, whichever is encountered at a lesser depth.

5.2 During the driling operations representative sample of the sediments shall be collected for every 6 meters or at the change of lithology whichever is earlier. The time taken for every 6 meters of drilling shall also be recorded.

5.3 The unit rate quoted by the tenderer shall be deemed to include collection and supplying the samples in suitable polythene Jars duly labelled as stated above to the entire satisfaction of GE and as directed by him.

5.4 200 mm dia casing pipes may be cut down to suitable lengths and welded to facilitated in lowering the casing and connecting to strainer and casing pipes. Four pieces of 300mm long of flats of size 50x6mm shall be welded on each joint. Contractor's quoted rate of relevant item is deemed inclusive of this provision.

5.5 The bail plug shall be provided at the bottom of the bore well casing and fixed securely. MS Screwed caps shall be provided at the top of the casing pipe 750mm above the ground level.

5.6 The MS casing pipe, piug and strainer pipe etc shall be given outside two coats of approved bitumenous anticorrosive paint before lowering in the bores. The surface of the pipe shall be thoroughly cleaned with wire brush and made free from rust before painting. The Engineer in-Charge shall approve the surface before paint is applied.

PARTICULAR SPECIFICATIONS(CONTD……)

6. **DEVELOPMENT OF WELLS**

The contractor shall develop the well by compressor to ensure that the well is fully developed by developing the well for a period as specified in Sch 'A' normally The discharge during development shall be measured at an interval of one hour and record thereof shall be kept by the contractor. The exact time of development shall be noted and paid under respective items of Sch 'A. The development process shall be continued as specified in Sch 'A' and if so ordered by the GE taking into consideration the following:

(i) Discharge ceases to improve at a fixed depression and alternatively depression ceases to improve at constant discharge.

(ii) The Water is free of sand and suspended matter. The water shall be considered sand free if the sand content or suspended matter are not more than five parts in million parts of water by volume within ten minutes of the start of pumping.

(iii) The discharge will be measured with a suitable 90º 'V' notch. An approved channel for the flow of water shall be provided over the ground with suitable baffles to eliminate the turbulence completely before the water is allowed to flow over the 'V' notch. The rear level and working level of water shall be recorded before commencement and during progress of well development.

7. **VERTICALITY TEST**

The entire assembly shall be installed in true vertical position. The verticality of the well shall be tested upto a depth of 5 metre above from the bottom of bore well with latest Engineer practice with plump bob, steel circular gauge etc. Tolerance to confirm to Provisions as per latest IS. Further verticality of bore well shall suit recommendation of pump manufacturers. The tube well not conforming to this specification shall be rejected.

8. **YIELD TESTS**

The yield test of the tube well shall be carried out by pumping out well water, after the well is satisfactorily developed. Pumping rate shall be increased gradually until the desired draw down is attained. After this the test shall be continued at the same draw down for 10 hours. The duration shall be increased suitably and upto 12 hours at the specific direction of the Engineer-in-Charge. During this period the contractor shall be required to collect, at his own cost, the sample of water coming out of the well in 2 lire capacity polyethylene bottles. For chemical and other analysis. The contractor shall be required to handover to the Engineer in-Charge one such sample for every well drilled. After the test, a certificate shall have to be signed by both the contractor and the Engineer-in-Charge about the yield of the well at the specified draw dawn.

9. **STRATE SAMPLES & RECORD GENERAL**

9.1 The contractor shall be deemed to have himself fully acquainted with the scope of the proposed works nature of soil and all matters affecting the execution of work. No claim whatsoever shall be entertained for want of knowledge on part of contractor to do so.

9.2 All work shall be carried out in accordance with the latest Engineering practice to the entire Satisfaction of the GE.

9.3 The contractor shall supply to the Engineer-in-Charge on completion of the tube wells a complete chart for the tube well in a standard tracing cloth showing the following without any extra payment for the same:-

(a) Strata change encountered with description and depths with particular reference to size and grain of sand and the water bearing strata.
(b) Spring level below ground level (aquifers with water bracing strata) opposite which strainer and slotted pipes have been located.
(c) Strata against which blind plain pipe have been located.
(d) Position of joints in slotted/blind pipes/housing pipes.
(e) Diameter of the bores and reduced sizes with depth if any.

PARTICULAR SPECIFICATIONS(CONTD……)

(f) Diameter and length of each section of bore well casing and strainer prevalence of sand etc.

(g) Yield test and duration of development operation and working level, prevalence of sand etc.

10. **PIPES AND FITTINGS**
(SAMPLES AND MATERIALS FOR THE WELLS)

10.1 All materials and fittings used for this work shall conform to latest IS/BS specifications as applicable.

10.2 The casing pipe shall be of plain iron welded pipe, thickness as specified in Sch "A", welded in standard length conforming to clause 7.3 above longitudinally and circumferencially with suitable sockets.

10.3 **SLOTTED/STRAINER**

10.3.1 These shall be made out of MS plate thickness and slots cut as specified in Sch "A"

10.3.2 The slot shall be or 1/16" wide and 2" long as ordered. The numbers of slots per 25mm of circumference and vertical spacing between the slots shall be all as decided and directed by the GE.

10.3.3 The slotted pipes shall be of standard length and oonnected to the 200mm dia plain MS pipe by suitable welds. The rates for the slotted pipe quoted by the contractor shall be deemed to include the cost of welds and nothing extra shall be adnissible to the contractor on this account.

11. **ABANDONMENT OF BORE WELLS**

The bore well shall be abandoned due to carving scouring during drilling or lowering the assembly or packing gravel. No payment whatsoever will be made to the contractor on this account.

12. **FILLING OF BORE WELLS**

The abandoned bore hole shall be filled with spoil, obtained from the drilling of this bore hole and compacted thoroughly all as directed by the Engineer-in-Charge and upto his entire satisfaction.

13. **TENATIVE LAYOUT FOR LOWERING OF CASING STRAINER PIPES/ASSEMBLY IN BORE HOLE**

Before lowering length of pipe in the tube well, the contractor shall submit a tentative layout plan on a paper with his recommendations for location of MS pipe and strainers, bail plug and the casing pipes which in his opinion would give the desired yield and obtained approval of the Engineer-in-Charge/GE in writing which shall be the basis of the final drawing on tracing cloth as specified in above clause which shall be signed by both the contractor and the Engineer-in-Charge,

14. **COMPLETION CERTIFICATE**

The work under this contract shall not deemed to be completed until the verticality and yield tests are carried out to the entire satisfaction of the GE and correctness are certified by him.

15. **GALVANIZED IRON PIPE**

15.1 GI pipes and fittings shall be as per Schedule 'A (BOQ) and shall comply with the requirements laid down in para 18.4 to 18.4.5 of MES Schedule (Part-I).

15.2 All materials including fittings, accessories etc shall be of approved make and shall comply with appropriate Indian Standards and ISI marked. Mild steel galvanised tubes (pipes) and fittings shall bear ISI certification mark with IS-1239 (Part-) and IS-1239 (Part-I) all as specified in clause 18.4 of MES SSR-2009 (Part-I).

PARTICULAR SPECIFICATIONS(CONTD……)

15.3 Contractor shall provide screwed plugs to all open end of pipe on completion of works without any extra cost to the department.

15.4 **LAYING AND JOINTING OF GI PIPES AND FITTINGS**

15.5 Laying and jointing of GI pipes shall be done as per para 18.50.1 to 18.50.4 and 18.51.1 to 18.51.5 of MES Schedule (Part-I).

15.6 **TESTING OF MAINS AFTER LAYING OF PIPES:** After laying the pipes, testing shall be carried out as per provisions of clause 18.54 of MES Schedule (Part-I).

15.7 **TESTING OF SERVICE PIPES & FITTINGS :** Refer Clause 18.55 on page No 378 of MES Schedule Part-I.

16. **SUBMERSIBLE PUMP SET**

16.1 The pump set shall be complete with submersible pump and a submersible motor with shafts connected by a sleeve and shall operate submerged beneath the surface of water. The pump shall be multistage centrifugal pump with radial or mixed flow impeller. The casing of radial flow impeller pumps shall be clamped together by flat steel hook bolts where as that of mixed flow impeller pumps by studs. Suction casing shall be between pump and motor and shall consist of non return valve at discharge end.

16.2 Submersible motor shall be water filled and water lubricated squirrel cage type. The axial thrust generated by the pumps should be absorbed by a thrust bearing fitted at the bottom of the motor. The diaphragm beneath the thrust bearing should be suitable to compensate the over pressure which arises as result of the thermal expansion of the filled water when the temperature of winding rises.

16.3 The pump shall be provided with sleeve bearings and the motor with sleeve and thrust bearing. All the bearings shall be water lubricated and protected to a large degree against the ingress of sand by suitable structural elements.

16.4 The motor shall be sealed by radial seal rings to avoid mixing of well water and the filled water.

16.5 Impellers shall be of gun metal graded material.

16.6 The selection of model of pumping set shall be done so as to produce the required BHP and discharge at required head as given in Sch A'. In case of difficulty in selection of model it shall selected in such a way that none of parameters are less than given in Sch A'. However no extra payment shall be done for higher capacity of pumping set in any of the parameter.

17. **CATALOGUES**

The contractor shall submit 3 copies of manufacturers illustrative technical catalogues separately indicating exhaustive instructions for installation, maintenance operation and also spare parts numbers of diagrams for each type of equipment within seven days from the date of acceptance.

18. **SUBMISSION OF MANUFACTURER'S TEST CERTIFICATE**

The Contractor shall submit manufacturers test certificate for equipments etc to GE without any extra cost.

PARTICULAR SPECIFICATIONS(CONTD……)

19. **GUARANTEE**

The tenderer shall guarantee all items of equipment including electrical machinery and wiring for a period of 12 (twelve) months (or period of guarantee as given by manufacturer whichever is more) after completion/installation and taking over by the GE.

20. **TESTING**

20.1 After the installation is ready in all respects 8 hours duration test shall be conducted by the Contractor before installation is finally taken over by the GE. Testing procedure shall be as per relevant IS and test details shall be recorded and signed by both the parties.

20.2 If the test results are not satisfactory, contractor shall rectify/replace the defective installation or any part thereof as directed by the GE at his own expense. Installation shall be taken over only after satisfactory test results of the installation. Work shall be treated as completed after such satisfactory testing.

20.3 All materials labour equipment etc required for carrying out testing of the installation shall be arranged by the contractor without any extra cost to department.

21. **DISINFECTION OF PIPING SYSTEM AND STERLIZATION OF MAIN:** Disinfection of piping system and sterilization of main shall be carried out as per Para 18.114.6 to 18.114.7 of MES Schedule (Part-I).

PARTICULAR SPECIFICATIONS(CONTD……)

FOLLOWING MAKES OF MATERIALS TO BE INCORPORATED

1. GI PIPES
(I) Tata (ii) Swastic (iii) Jindal. (iv) Parkash

2. GI PIPE FITTINGS
(i) NVR (ii) R (iii) SVW (iv) MAC

3. C.I PIPES
(i) Tata (ii) Electro Steel (iii) Kesoram (iv) Kejrewal

4. MS PIPES
(i) Tata (ii) Swastic (iii) Jindal (iv) Prakash (v) BST

5. DI PIPES
(i) TATA (ii) Lanco (iii) Jindal Saw (iv) Electro Therm

6. SUBMERSIBLE CABLE
(i) Havells (ii) BCH (iii) RR Kable (iv) Finolex
(v) Polycab (vi) HPL (vii) Bonton

7. SLUICE VALVE

(i) Kirloskar (ii) Sant (iii) Bir (iv) Varun
(v) Venus (vi) NVR (vii) Leader (viii) Upadhyay (ix) Kartar

8. REFLUX VALVE/NON RETURN VALVE

(i) Bir Varun (ii) Varun (iii) Venus (iv) NVR

(v) Kirloskar (vi) Kartar

9. PVC INSULATED COPPER CONDUCTOR CABLE

(i) Finolex (ii) Plaza (iii) Havells (iv) Indo Asian

10. Submersible Pumps/Motors
(i) Kirlosakar (ii) Jyoti (iii) Calama (iv) KSB (v) Great India (vi) BS
(vii) Crompton (viii) Beacon

11. Centrifugal Pumps/Motors
(i) Johnson (ii) Calama (iii) Kirloskar (iv) Jyoti (v) Mather and Platt (vi) Beacon

(vii) KSB (viii) Crompton

12. Air Release Valve

(i) Bir (ii) Venus (iii) Kirloskar (iv) Leader

13. Gate Valve

(i) BVI (ii) BS (iii) NVR (iv) ZOLOTO (v) SANT

Signature of Contractor

AGE (Contracts)
For Accepting Officer

APPENDIX 'C'

ANNEXURE TO SPECIAL CONDITIONS
TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES

Sl No	Materials	Test	Method of testing	Frequency of tests			Level of tests	Rate per test Rs.	Remarks
1	2	3	4	5			6	7	8
1.	Bricks (Except Leh and Kargil area)	(i) Compressive Strength	IS-3495 (Part-II)	As per IS-5454 as given under :-			'A'	330/-	Checks for visual and dimensional characteristics shall also be carried out as per IS-5454
		(ii) Water absorption	-do-	Lot size	Sample Size	330/-	'A'	330./-	Legend **'A'-Zonal lab [Site lab (established at site by the contractor)]** **'B'-Zonal lab**
		(iii) Efflorescence	-do- (Part-I)	1001 to 10000	5	330/-	'A'	330	**'C'-National test house/SEMT wing Pune/Engg College.**
				10001 to 35000	10				
				35001 to 50000	15				
2.	Coarse aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of aggregates or part thereof brought to site			'A'	660/-	
		(ii) Flakiness Index	-do-	-do-			'A'	250/-	
		(iii) Estimation of deleterious materials	IS-2386 (Part-I)	One test for every 100 Cum of aggregates or part thereof.			'A'	600/-	
		(iv) Organic impurities	IS-2386 (Part-I)	One test per source of supply			'C'	275/-	

Contd …/

APPENDIX 'C' Contd….

ANNEXURE TO SPECIAL CONDITIONS

TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES-

1	2	3	4	5	6	7	8
2. (Contd)	Coarse aggregate	(v) Moisture content	IS-2386 (Part-II)	Regularly as required	'A'	330/-	
		(vi) Specific gravity	IS-2386 (Part-II)	One test for each source of supply	'B'	330/-	
3.	Fine aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15 Cum of fine aggregate or part when brought to site	'A'	180/-	
		(ii) Test for clay, silt and impurities	IS-2386 (Part-II)	-do-	'A'	500/-	
		(iii) Specific Gravity	IS-2386 (Part-II)	One for each source of supply	'B'	330/-	
		(iv) Moisture content	IS-2386 (Part-II)	Regularly as required subject to 2 tests per day when being used	'A'	330/-	
		(v) Test for organic impurities	IS-2386 (Part-II)	One test for each source of supply	'C'	275/-	
4	Cement	(i) Setting time	IS-4031-63 affirmed 1980	One for each consignments or as when required	'B'	500/-	
		(ii) Soundness	-do-	-do-	'B'	550/-	
		(iii) Compressive strength	-do-	-do-	'B'	550/-	
		(iv) Fineness	-do-	-do-	'B'	275/-	

Contd…/-

APPENDIX 'C' Contd….

ANNEXURE TO SPECIAL CONDITIONS

TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES

1	2	3	4	5		6	7	8
5.	Structural Concrete (M-20 grade and above)	(i) Slump test (OR) compacting factor test vee-bee time test	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under :-		'A'	300/-	Random sampling shall be carried to cover all mix unit
		(ii) Compressive strength	IS-516	Qty of concrete **In the work** (Cubic metre)	No of Samples	'A'	900/-	
				1-5	1			
				6-15	2			
				16-30	3			
				31-50	4			
				51 and above	4+1 for each Addl 50 Cum or part thereof			
6.	(a) PCC Block for walling (Hollow Block)	(i) Compressive strength	IS:2156-1984 (Appx 'B')	08 Blocks out of 14		'A'	900/-	Sample :- 14 blocks from consignment of every 5000 blocks or part thereof.
		(ii) Water absorption	-do- (Appx 'E')	03 Blocks out of 14		'B'	330/-	
		(iii) Density	-do- (Appx 'A')	03 Blocks out of 14		'B'	330/-	
	(b) PCC solid Block for walling	(i) Compressive strength	IS-2185	12 Blocks out of 18		'A'	900/-	Sample :- 18 blocks from consignment of every 1000 blocks or part thereof.
		(ii) Water absorption	-do-	03 Blocks out of 18		'B'	330/-	These blocks to be checked for dimension and weight
		(iii) Density	-do-	03 Blocks out of 18		'B'	330/-	

Contd…/-

APPENDIX 'C' Contd….

ANNEXURE TO SPECIAL CONDITIONS

TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES

1	2	3	4	5	6	7	8
7.	Cement Flooring Tiles/ Terrazzo Tiles	(i) Water absorption	IS-1237 (Appx 'D')	06 tiles out of 18	'B'	330/-	Sample of 18 tiles from each source of supply selected at random.
		(ii) Wet transverse strength	IS-1237 (Appx 'E')	-do-	'B'	660/-	
		(iii) Resistance to wear	IS-1237 (Appx 'F')	-do-	'C'	1000/-	
8.	Burnt clay roofing tiles (hand made) As per IS-2690 (Part-II)	(i) Water Absorption	IS-3495 (Part-II)	6 tiles out of 12	'B'	216	Samples :- 12 tiles from each source of supply selected at random
	Length 150mm to 250mm	(ii) Compressive strength	IS:3495 (Part-i)	-do-	'A'	180/-	
	Width 100mm to 200mm						
	Thickness 35mm to 50mm						
9.	Mangalore pattern roofing tiles	(i) Water Absorption	IS-654 (Appx 'A')	6 tiles out of 32	'B'	180/-	} Samples :- 32 tiles from each consignment
		(ii) Breaking Load	IS-654 (Appx 'C')	-do-	'B'	120/-	} of 3000 tiles or part thereof. These tiles shall be checked for dimension.
10.	TIMBER	(i) Specific gravity and weight	IS-1708-1960	Minimum 3 samples from a lot of 4 Cum or 250 pieces of seasoned timber	'B'	120/-	
		(ii) Moisture Content	-do-	-	'A'	120/-	

Contd…/-

APPENDIX 'C' Contd….

ANNEXURE TO SPECIAL CONDITIONS

TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES

1	2	3	4	5	6	7	8
11.	Water for construction purposes	(i) Test for acidity	IS-456 & 3015	Once at the stage of approval of source of water	'B'	500/-	Also refer clause 4.3 of IS0456 and its subsequent sub clauses regarding suitability of water.
		(ii) Test for alkalinity	-do-	-	'B'	500/-	
		(iii) Test for solid content	-do-	-	'C'	500/-	
12.	Welding of Steel Work	Visual inspection test	IS-822-1970	100% by visual inspection	Work Site	360/-	Specialised tests, their method and frequency to be decided on consideration of their importance by the Accepting Officer.
13.	Timber panelled and glazed door/window & shutters (including factory made shutters)	(a) Dimensions, sizes, workmanship and finish	IS-1003-1977 (Part-I)	Frequency of sampling from each lot shall be as under :- Lot Size / Sample Size 26 to 50 / 5 51 to 100 / 8 101 to 150 / 13 151 to 300 / 20 301 to 500 / 32 5001 to 1000 / 50 1001 and above / 80	'A'	180/-	
		(b) Strength test (i) Slamming	IS-1303-1990	From the each lot 5% of the factory made shutters shall be manufacturer tested for strength tests.			
		(ii) Impact indentation	-do-				
		(iii) Shock resistance	-do-				
		(iv) Edge loading	-do-				

APPENDIX 'C' Contd….

ANNEXURE TO SPECIAL CONDITIONS
TEST OF MATERIALS AND RECOVERY RATES OF TESTING CHARGES

1	2	3	4	5	6	7	8
14.	Ply Wood (IS-303-1989)	(a) Moisture content	IS:1734-1983 (Part-I)	Six tests of pieces cut from each of the boards selected shall be subjected to tests.	'C'	240/-	Sampling shall be as per IS-7835-1975 tables
15.	Wood particles board (medium density) (IS-3097-1985)	(a) Density	IS-2360 (Part-III)	Three test specimen from each sample (size 150mm x 75mm)	'A'	60/-	Sampling shall be as per IS-3487-83 with moisture metre
		(b) Moisture content	-do-	-do-	'A' & 'B'	60/-	
		(c) Water Absorption	-do- (Part-16)	-do- (size 300mm x 300mm)	'A'	60/-	
		(d) Swelling due to surface absorption	-do- (Part-17)	-do- (size 125mm x 100mm)	'A'	60/-	
		(e) Swelling in water	-do-	-do- (size 200mm x 100mm)	'A'	60/-	
		(f) Modules of rupture	-do- (Part-4)	Three test specimens as per IS-2380-77	'B'	90/-	
		(g) Screw withdrawal strength	-do- (Part-4)	-do- as per IS-2385	'C'	120/-	

Signature of Contractor

AGE(Contracts)
for Accepting Officer

Notes :1. The cost of test other than these mentioned above which are required as per contract provision or specified in relevant IS/SSR/IES shall be borne by the contractor 2. The required number of blocks for testing shall be taken from every consignment of 5000 blocks. The block shall be picked up at random from a stack of particular batch. The sample of block shall be marked for future identification of the batch it represents

APPENDIX 'D'

CEMENT SUPPLY/ACCEPTANCE REGISTER

1. CA No. & Name of Work:
2. Control No :*
3. Name of Manufacturer/Brand Name /Gde of Cement (a) Manufacture_____(b) Brand_____(c)Grade_____
4. Qty of cement & lot No/Week No(in Bags) : (a) Qty_____ (b) lot No/ Week No._____
5. Manufacturer's test Certificate No. _____
6. Random Test details (a) Physical test report from _____vide their letter No._____ (Name of approval Lab /Engg College)

(b Chemical test report from _____vide their letter No._____(Name of approval Lab /Engg College)

7. Details of Physical & Chemical Properties.

Srl No	Details	Physical requirements (As per IS 4031)									Chemical requirements (As per IS 4032)								
		Specific Surface area (M2/Kg)	Soundness by Le Chatellear	Soundness by Auto Clave	Initial setting time (Minutes)	Final setting time (minutes)	Compressive Strengths (Mpa)			Temp during testing .C)	Standard Consistency (%)	Lime saturation factor (Ratio)	Alumina Iron Ration (Ratio)	Insoluble residue(%)	Magnesium (%)	Sulphuric Anhydride(%)	Loss on Ignition (%)	Alkalis (%)	Chlorines(%)
							03 days (Mpa)	07 days (Mpa)	28 days (Mpa)										
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	As per relevant IS																		
2	As per manufacture's test certificate																		
3	As per random test certificate																		

Remarks with signature
Accepted /Rejected ……Contractor Junior Engineer Engineer in-charge Garrison Engineer
Remarks of BOO/Inspecting Officer/CWE
*To be allotted serially by GE consignment wise_____

AGE(Contracts)
(Signature of the Contractor) For Accepting Officer

APPENDIX 'E'

STEEL SUPPLY/ACCEPTANCE FORM

1. CA No. & Name of Work:
2. Control No.
3. Particulars of Manufacturer TC No.
4. Details of Supplier (if any)
5. Random Test details (a) Physical test report from _____vide their letter No._____(Name of approval Lab /Engg College)

   (b) Chemical test report from _____vide their letter No._____ (Name of approval Lab /Engg College)

Srl No	Nomenclature and size of steel	IS Ref	Quality		Physical Properties								Remark
			Actual weight	Conversion Weight	Proof Stress/Yield Stress (N/mm2) 0.2% proof			Elongation %					
					IS	Test Sheet	Random Test	IS	Test Sheet	Random Test	Bend Test	Re-bend Test	
1	2	3.	4.	5	6	7	8	9	10	11	12	13	14

Remarks with signature Accepted /Rejected

Contractor Junior Engineer Engineer in-charge Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

_____
(Signature of the Contractor)

AGE(Contracts)
For Accepting Officer

GENERAL CONDITIONS OF CONTRACTS (IAFW-2249-1989 PRINT)
MEASUREMENT CONTRACTS (1779-A)

A COPY OF THE GENERAL CONDITIONS OF CONTRACTS-IAFW-2249(1989 Print) with Errata and amendment has been supplied to me/us and is in my/our possession. I/we have read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/We shall abide by the terms and conditions thereof.

It is hereby further agreed and declared by me/us, that the GENERAL CONDITIONS OF CONTRACTS-IAFW-2249 (1989 Print) including Condition 71 thereof pertaining to settlement of disputes by Arbitration containing 40 pages (serial Page Nos to with errata 1 to 20 and amendment Nos 1 to 49 form part of these tender documents.

AGE (Contracts) CONTRACTOR
FOR ACCEPTING OFFICER

SCHEDULE OF MINIMUM FAIR WAGES

1. It is hereby agreed that the "Schedule of Minimum Wages" as published vide Government of India vide Notification dated 20 May 2009 (comprising 9 pages) as amended subsequently till the date of issue of this tender shall be the "Fair Wages" under condition 58 of IAFW-2249 and shall forms part of these tender documents.

My/Our signature hereunder amounts to my/our having read and understood the provisions contained therein and I/We agree that I/We shall abide by the same and that aforesaid documents form part of this tender.

NOTE :1. "Schedule of Minimum Fair Wages" referred to above is available for reference, in any MES Office at under HQ 31 Zone.

2. The Schedule of Minimum Fair Wages as enclosed is for reference only. However, minimum Fair Wages mean the minimum wages of laborers as fixed by Govt of India and State Govt, whichever is higher.

AGE (Contracts)
FOR ACCEPTING OFFICER

CONTRACTOR